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Review article

Accumulation of perfluorinated alkyl substances (PFAS) in agricultural plants: A review

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ABSTRACT

PFASs are a class of compounds that include perfluoroalkyl and polyfluoroalkyl substances, some of the most persistent pollutants still allowed - or only partially restricted - in several product fabrications and industrial applications worldwide. PFASs have been shown to interact with blood proteins and are suspected of causing a number of pathological responses, including cancer. Given this threat to living organisms, we carried out a broad review of possible sources of PFASs and their potential accumulation in agricultural plants, from where they can transfer to humans through the food chain. Analysis of the literature indicates a direct correlation between PFAS concentrations in soil and bioaccumulation in plants. Furthermore, plant uptake largely changes with chain length, functional group, plant species and organ. Low accumulations of perfluorooctanoic acid (PFOA) and perfluorooctane sulfonic acid (PFOS) have been found in peeled potatoes and cereal seeds, while short-chain compounds can accumulate at high levels in leafy vegetables and fruits. Significant variations in PFAS buildup in plants according to soil amendment are also found, suggesting a particular interaction with soil organic matter. Here, we identify a series of challenges that PFASs pose to the development of a safe agriculture for future generations.

1. Introduction

Contamination of regional, rural and urban ecosystems by anthropic activities has been a constant concern in environmental sciences in recent decades, and environmental regulations have to be continuously updated as new pollutants emerge (Manzetti et al., 2014). Poly- and perfluorinated alkyl substances (PFASs), especially perfluoroalkyl acids (PFAAs), which include perfluoroalkyl carboxylic acid (PFCAs) and perfluoroalkyl sulfonic acids (PFSAs), are among the world's major pollutants. PFASs have been detected in oceans, across continents, and in remote parts of the globe, including the North Pole (Bossi et al., 2005; Cai et al., 2011; Holmström et al., 2005; Paul et al., 2008; Zhao et al., 2012), with several ecosystems in the USA, China and Europe affected (Ahrens et al., 2010; Herzke et al., 2013; Jin et al., 2009; Kowalczyk et al., 2013; Loos et al., 2009; So et al., 2004, 2007; Washington et al., 2010).

PFASs are a family of molecules composed of a carbon chain that can be linear or branched, and fully or partially fluorinated (Buck et al., 2011). They have unique physicochemical characteristics due to the larger size of the fluorine atom with respect to the hydrogen atom, and the strength of the carbon-fluorine bond, which give fluoroalkyl moieties their high thermal, chemical and biochemical stability. (Krafft and Riess, 2015; O'Hagan, 2008). This group of molecules also includes compounds with low aqueous surface tension and the property of interacting with water and non-polar phases, and they therefore exhibit amphiphilic behavior. These features have been widely exploited in several industrial and commercial applications for the preparation of fluorinated polymers and surfactants since 1950s. Examples include oil-repellent coatings for food containers and cookware, firefighting foams, coloring and waterproofing agents for fabrics, detergents, floor waxes, and pesticide formulations (Buck et al., 2011; Kissa, 1994). As a result of their wide use and high persistence, a broad range of these

Abbreviations: PFASs, Poly- and Perfluoroalkyl substances; PFAAs, Perfluoroalkyl acids; PFCAs, Perfluoroalkyl carboxylic acids; PFSAs, Perfluoroalkyl sulfonic acids; PFBA, Perfluorobutanoic acid; PFPeA, Perfluoropentanoic acid; PFHxA, Perfluorohexanoic acid; PFHpA, Perfluoroheptanoic acid; PFOA, Perfluorooctanoic acid; PFNA, Perfluorononanoic acid; PFDA, Perfluorodecanoic acid; PFUnA, Perfluoroundecanoic acid; PFDoA, Perfluorododecanoic acid; PFBS, Perfluorobutane sulfonic acid; PFHxS, Perfluorohexane sulfonic acid; PFOS, Perfluorooctane sulfonic acid

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substances have been detected in the environment, and in wildlife and humans. PFASs accumulate in the body, mainly in the liver, with mean half-life values of 3.8 years for PFOA and 5.4 years for PFOS, though with considerable variability (EFSA, 2008). They saturate protein surfaces (Bischel et al., 2010; Chen and Guo, 2009; Manzetti et al., 2014; Qin et al., 2010), and are transferred through the maternal cord from the placenta in rat models as well as in humans (Inoue et al., 2004; Loccisano et al., 2012; Winkens et al., 2017). Studies on rats showed that PFOA and PFOS accumulate mainly in liver, kidneys and serum (EFSA, 2008).

The toxicity of PFASs, particularly PFOA and PFOS, has been extensively studied with clear results on animals, but less statistical evidence regarding humans (for details, see individual chapters in DeWitt, 2015). Less is known regarding other compounds, including those with a short chain, such as perfluorobutanoic acid (PFBA), perfluorohexanoic acid (PFHxA) and perfluorobutane sulfonic acid (PFBS), which have been recently introduced by the manufacturing industry following concerns about the undesired effects of PFOA and PFOS on human health and the environment (Ritter, 2010). Although some short-chain PFASs, i.e. $C < 7$ for PFCAs and $C < 6$ for PFSAs (Buck et al., 2011), are not considered bioaccumulative, they are as recalcitrant to degradation as long-chained PFASs, are highly soluble, and have lower sorption to solids, which lead to greater mobility in the environment (Vierke et al., 2012; Wang et al., 2015)

Adverse effects associated with PFAS exposure in animal models include hepatotoxicity, tumor induction, developmental toxicity, immunotoxicity, neurotoxicity and endocrine disruption (Table 1; DeWitt, 2015 for a recent review). Regarding human health, the most comprehensive epidemiological data linking PFOA exposure and health outcomes were gathered by the C8 Science Panel (http://www.c8sciencepanel.org), which studied Mid-Ohio Valley communities that had been potentially affected by releases of PFOA since the 1950s. Probable links between serum levels and several health endpoints, such as increased levels of cholesterol, liver enzyme activity and uric acid, altered thyroid parameters, and pregnancy-induced hypertension, have emerged from these studies. The same investigations also report a probable link between PFOA exposure and testicular and kidney cancers (http://www.c8sciencepanel.org). The International Agency for Research on Cancer (IARC, 2017) has recently classified PFOA as possibly carcinogenic, while the US Environmental Protection Agency (EPA) concluded that both PFOA and PFOS are possibly carcinogenic to humans.

Among the most common PFASs found in the environment (Table 1), PFOA and PFOS are considered some of the most widespread organic pollutants for biota and humans (So et al., 2004; Ahrens, 2011; Houde et al., 2011; Kannan et al., 2001, 2005; Yamashita et al., 2005). However, other PFASs have been detected in the environment, such as PFBA, perfluoroundecanoic acid (PFUnA), and perfluorododecanoic acid (PFDoA) (Ahrens et al., 2009). Their persistence is strongly associated with carbon chain length, and PFASs with more than five ($C > 5$) carbon atoms are biopersistent and may have higher K_{ow} values than those with shorter chain lengths (Eriksen et al., 2010; Latała et al., 2009; Parsons et al., 2008). Reactivity, on the other hand, is less dependent on chain length, but is instead related to the non-fluorinated part of the molecules and/or the functional groups of the PFAS end-region (i.e. the sulfonic or carboxylic groups). The fluorinated part can generally be regarded as inactive, given the high strength of the carbon-fluoride bond, 109–130 kcal/mol (O'Hagan, 2008; Fuchibe and Akiyama, 2006). It is important to underline that the substances containing a perfluoroalkyl moiety have the potential to be transformed abiotically or biotically into perfluoroalkyl substances through changes in the non-fluorinated part of the molecule (Buck et al., 2011). Therefore, the corresponding carboxylic acids and sulfonic acids are formed from several precursors, such as fluorotelomer alcohols, sulfonamide derivatives, and polyfluoroalkyl phosphates. For this reason, we focus our review on the uptake and distribution in plants of PFAAs (PFCAs and PFSA) rather than their precursors.

2. Sources of PFASs in humans

Water and food consumption are generally considered the two major sources of PFASs in humans, although air and air-suspended dust, food packaging and cookware also contribute to the overall PFAS load in the human body (Tittlemier et al., 2007). Seafood, especially when fished in hotspot waters with much higher PFAS concentrations than the background, is considered one of the most important food sources of PFASs in humans (Brambilla et al., 2015; Christensen et al., 2017; D'Hollander et al., 2010a, 2010b; Fromme et al., 2009; Hloušková et al., 2013).

Herzke et al. (2013) and D'Hollander et al. (2015) recently reported the presence of PFASs, particularly PFCAs, in various vegetables, cereals and fruits sampled in Belgium, the Czech Republic, Italy and Norway. Furthermore, a series of scientific articles on PFAS accumulation in plants is emerging across the world (vide infra). Plants may, therefore, be considered possible contributors to the PFAS body burden in humans, either directly as part of the human diet (Klenow et al., 2013) or indirectly as fodder for livestock (Kowalczyk et al., 2013).

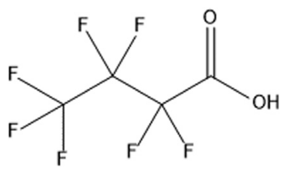
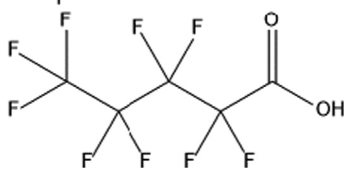
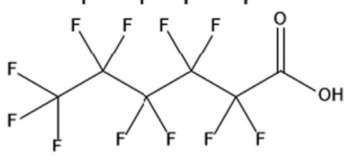
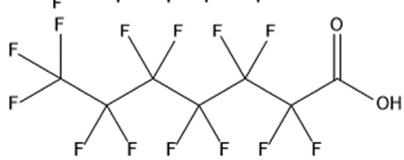
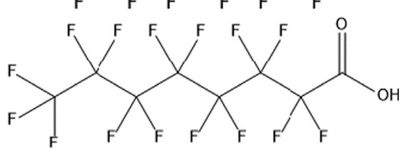
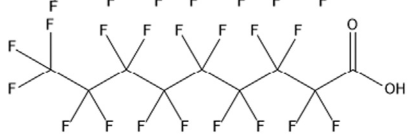
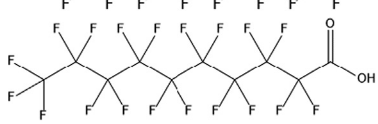
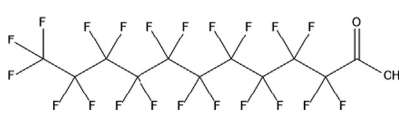
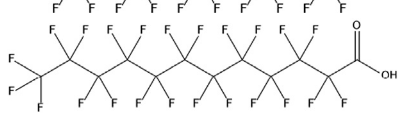
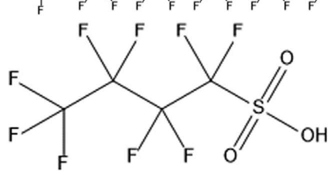
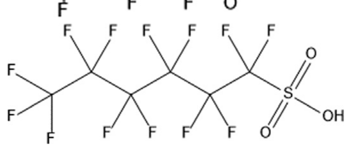
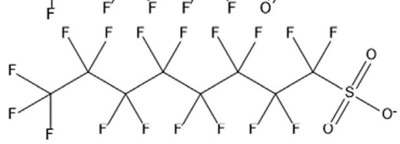
The purpose of this paper is to review the recent literature regarding the uptake and partitioning of the most common PFASs in agricultural plants, taking into account both cereals and vegetables, in an attempt to draw a picture of the role of the biotic and abiotic factors affecting these processes.

3. Sources of PFASs in plants

Due to the partial water solubility of PFASs, water is assumed to be their main vehicle of transfer across environmental compartments and biota, making a substantial contribution to their diffusion. High concentrations of PFAAs, which encompass both PFCAs and PFSAs, are often reported in effluents from wastewater treatment plants (WWTPs) (Llorca et al., 2012; Xiao et al., 2012), indicating that both urban and, particularly, industrial purification plants contribute to PFAS delivery to the environment. It has also been shown that PFOS and other long-chained PFSAs and PFCAs are mobilized from landfills and released into water bodies (Busch et al., 2010; Huset et al., 2008; Weber et al., 2011). In addition, runoff and leaching from areas treated with firefighting foams make a considerable contribution to PFAS pollution (Ahrens, 2011; Backe et al., 2013; Brambilla et al., 2015; Filipovic et al., 2015; Korucu et al., 2015). PFOS levels ranging from < 10 ng L^{-1} to 100 ng L^{-1} have been reported in surface and ground waters (Brambilla et al., 2015), but PFOA concentrations at the μg L^{-1} level have been measured in surface waters in Italy and in the Ruhr region of Germany (Valsecchi et al., 2015; Skutlarek et al., 2006). Alarming values at the mg L^{-1} level in ground water and surface water close to fire-training locations and airports have also been measured (Ahrens, 2011; Moody and Field, 1999). Field irrigation with contaminated surface or ground waters may be an important source of soil and plant contamination by PFASs.

On agricultural land, a further important source of perfluorinated compounds comes from the use of contaminated sewage sludge as a soil conditioner (Sepulvado et al., 2011; Zareitalabad et al., 2013). In fact, PFASs entering WWTPs, although with differences due to their chain length and functional group, can partly escape into the effluent and partly sorb to sludge due to their amphiphilic behavior (Chen et al., 2013). In some areas of Germany, PFOS and PFOA have been measured in concentrations of 1000 and 910 μg/kg soil dry weight (d.w.), respectively, in the top 0.3 m layer of the soil, and in concentrations of 5500 and 810 μg/kg d.w., respectively, at 0.3–0.6 m, as a result of the application of soil amendments derived from industrial waste (Wilhelm et al., 2008, 2009). In the USA, the use of contaminated sludge as a soil conditioner has caused contamination ranging from 500 to 2000 μg/kg to 4000–6000 μg/kg d.w. of total PFASs, according to the year and field surveyed. Perfluorodecanoic acid (PFDA) was the major PFAS

Table 1
A list of the most common perfluorinated alkyl substances (PFASs) with their applications and properties.

Compound	Use	Structure	MW	Solubility in water	Bioaccumulative properties	Confirmed toxic effects
Perfluorobutanoic acid, PFBA	Synthetic chemistry		214.04	Soluble	No (Conder et al., 2008; Wang et al., 2015)	–
Perfluoropentanoic acid, PFPeA	Breakdown product of stain- and grease-proof coatings on food packaging, couches, carpets		264.05	–	No (Hölzer et al., 2008) Binds to serum (Qin et al., 2010).	Increase in liver sAST protein (Upham et al., 2009)
Perfluorohexanoic acid, PFHxA	Unknown		314.05	Miscible	Yes, Log K_{ow}: 4.37 (Zhao et al., 2013; Fan et al., 2014)	Yes Kidney accumulation (Klaunig et al., 2014)
Perfluoroheptanoic acid, PFHpA	Breakdown product of stain- and grease-proof coatings on food packaging, couches, carpets.		364.06	9.5 g/l	Yes (Haukås et al., 2007) Log K_{ow}: 5.33 (Zhao et al., 2013)	Yes (Rosen et al., 2013; Vongphachan et al., 2011; Wolf et al., 2012)
Perfluorooctanoic acid, PFOA	Water and oil repellant in fabrics and textiles, food packaging.		414.07	No	Log K_{OC}: 2.4 ± 0.12 (Ahrens et al., 2011b) Log K_D: − 0.22–0.55 (De Voogt and Sáez, 2006)	Yes (Abbott et al., 2007; Kudo and Kawashima, 2003; Wolf et al., 2007)
Perfluorononanoic acid, PFNA	Surfactant for synthesis of textiles and polymers		464.08	9.5 g/l	Yes (Henderson and Smith, 2007) Log K_{ow}: 6.3 (Zhao et al., 2013)	Yes (Henderson and Smith, 2007; Fang et al., 2008; Upham et al., 2009; Wolf et al., 2010)
Perfluorodecanoic acid, PFDA	Breakdown product of stain- and grease-proof coatings on food packaging, couches, carpets		514.08	No	Yes (Buck et al., 2011; Zhao et al., 2013; Jones et al., 2003; Martin et al., 2003a, 2003b) Log K_{ow}: 8.23 (Zhao et al., 2013)	Yes (Harris and Birnbaum, 1989; Heuvel et al., 1992)
Perfluoroundecanoic acid, PFUnA			564.09	9.5 g/l	Yes, Log K_{ow}: 9.20 (Zhao et al., 2013; Bjermo et al., 2013)	Yes (Takahashi et al., 2014)
Perfluorododecanoic acid, PFDoA			614.10	9.5 g/l	Yes, Log K_{ow}: 10.2 (Zhao et al., 2013)	Yes (Liu et al., 2016; Ren et al., 2015; Zhang et al., 2013)
Perfluorobutane sulfonic acid, PFBS	Stain repellant		300.10	1.4 µg/l (Martin et al. (2003a)	Yes (Jones et al., 2003; Martin et al., 2003b)	Yes (Hu et al., 2003)
Perfluorohexane sulfonic acid, PFHxS	Surfactant for textiles		400.11	–	Yes, Log K_{ow}: 4.34 (Wilhelm et al., 2009; Zhao et al., 2013; Calafat et al., 2006)	–
Perfluorooctanesulfonate, PFOS	Firefighting foam, textiles		500.13	680 mg/l (You et al., 2010)	Yes. Log K_{ow}: 6.28 (Houde et al., 2011; Zhao et al., 2013). Globally distributed in wild-life (Giesy and Kannan, 2001)	Yes (Ankley et al., 2005; Seacat et al., 2002, 2003)

(< 990 µg/kg d.w.), while PFOA was measured at 320 and PFOS at 410 µg/kg d.w. (Washington et al., 2010). The use of PFAS as an emulsifier in phytosanitary product formulations may also contribute to the presence of these substances in the above-ground edible parts of vegetables (Lassen et al., 2012). Furthermore, an alternative pathway of PFAS accumulation in plants is the aerial transport of volatile or particle-bound precursors, which can be absorbed by plant leaves with further metabolization to PFCAs and PFSAs (Rankin et al., 2016; Simonich and Hites, 1995; Yoo et al., 2011). Air emissions of PFASs from landfills and WWTPs (Ahrens et al., 2011a), and accumulation of neutral and ionizable PFASs in leaves around landfills (Tian et al., 2018) have been reported. The spread of PFASs via dust particles has been documented as reaching the polar regions (Schenker et al., 2008; Rankin et al., 2016). Lindim et al. (2016) estimated that the contribution of river discharge to PFOS input to the Baltic proper, the Black Sea and the Adriatic Sea was on average 3.5 times greater than that of atmospheric deposition, whereas atmospheric deposition of PFOA was similar to or larger than riverine disposal.

4. PFAS accumulation in plants

4.1. Cereals

As far as we know, the first systematic research on PFOA and PFOS transfer from soil to plants was conducted by Stahl et al. (2009). In their investigations on spring wheat, oats, potatoes, maize and perennial ryegrass grown in soil spiked with PFOA and PFOS in the range 0–50 mg/kg soil, they found that these pollutants accumulated in all the plant species, the tissue concentrations of PFOA being generally higher than those of PFOS. Accumulation was found to increase with soil pollutant concentrations, with values higher in green parts (vegetative) than in grains. Large differences were found among plant species, possibly due to their different protein contents, as indicated by Wen et al. (2016), or to the different compositions and surface areas of their root systems, or to biomass accumulation, as suggested by Müller et al. (2016). The amount of water transpired during growth has also been suggested to account for different uptakes and translocation capabilities among crops (Blaine et al., 2014a; 2014b). Therefore, all the climatic parameters influencing stomatal opening, such as irradiance, temperature and humidity, can influence plant uptake of perfluorinated compounds. In Stahl et al.'s (2009) experiments, PFAS concentrations in cereal straw generally followed the hierarchy maize < < oats < wheat, as shown in Table 2, which reports values obtained at a soil concentration of 1 mg/kg, a level comparable to those reported by Wilhelm et al. (2008) and Washington et al. (2010). Stahl et al. (2009) also showed that perennial ryegrass accumulated PFAS to different extents at four successive cuttings, with the highest concentrations at the second. Given the high level of PFAS accumulation (PFOA in particular), which ranged from 408 to 7520 µg/kg d.w. (Table 2), where the soil is contaminated this forage species can represent an entry point of PFASs into the food chain via animal feeding.

Krippner et al. (2015) deepened the study of PFAS accumulation in the straw and grains of maize plants by looking at ten C4 to C10 substances containing the carboxylic (PFCAs) or sulfonic (PFSAs) functional groups, all individually spiked at 0.25 and 1 mg/kg soil. They confirmed the direct correlation between soil and plant tissue concentrations: the higher the soil PFAS content, the more the plants accumulate. Differences in the accumulation levels of compounds in plants depend on chain length, with the longer-chained compounds generally having lower accumulation rates, and on the associated functional groups, with PFSAs generally having lower accumulation rates than PFCAs. Krippner et al. (2015) also confirmed straw as having a much greater rate of accumulation than kernels, the former having a total PFAS content of around 52,000 µg/kg d.w., the latter around 900 µg/kg d.w. at the individual spiking dose of 1 mg/kg soil for a total of 10 mg/kg soil (Table 2). At the same spiking dose, the highest

Table 2
Initial PFAS concentrations (mg/kg soil) and their accumulation in different crop plant parts (µg/kg d.w.).

Plant species and organ	Compounds	Type of treatment and initial soil concentrations (mg/kg soil)	Concentrations in plants (µg/kg d.w.)	References
Maize straw	PFOA, PFOS	Spiked soil. 0–50 for each compound	At 1 mg/kg: PFOA: 126 PFOS: 104	Stahl et al. (2009)
Maize straw	PFAA mixture, ten compounds	Spiked soil. 0.25 and 1 for each substance	At 1 mg/kg: PFBA: 35233 PFBS: 1843 PFOA: 645 PFOS: 617 ΣPFASs = 52200	Krippner et al. (2015)
Maize stover	PFAA mixture	Soil + urban 2x biosolids. 0.00716, as Σ. PFBA: 0.00010 PFBS: 0.00039 PFOA: 0.00128 PFOS: 0.00282	PFBA: 4.21 PFBS: < LOQ PFOA: < LOQ PFOS: < LOQ ΣPFASs = 4.49	Blaine et al. (2013)
Maize leaves	PFOS	Spiked soil. 38.5 PFBS as by-product: 0.020	PFOS: 23100 PFBS: 120	Navarro et al. (2017)
Maize ears	PFOA, PFOS	Spiked soil. 0–50 for each compound	At 1 mg/kg soil: PFOA: 4 PFOS: 3	Stahl et al. (2009)
Maize kernels	PFAA mixture, ten compounds	Spiked soil. 0.25 and 1 for each substance	At 1 mg/kg: PFBA: 229 PFBS: 5.44 PFOA: 2.45 PFOS: < LOQ ΣPFASs = 860	Krippner et al. (2015)
Maize roots	PFOS	Spiked soil. 38.5	254000	Navarro et al. (2017)
Oat straw	PFOA, PFOS	Spiked soil. 0–50 for each compound	At 1 mg/kg: PFOA: 690 PFOS: 150	Stahl et al. (2009)
Oat grains	PFOA, PFOS	Spiked soil. 0–50 for each compound	At 1 mg/k: PFOA: 54 PFOS: 17	Stahl et al. (2009)
Ryegrass, four cuttings	PFOA, PFOS	Spiked soil. 0–50 for each compound	Range at 1 mg/kg within the four cuttings: PFOA: 408–7520 PFOS: 92–470	Stahl et al. (2009)
Wheat straw	PFOA, PFOS	Spiked soil. 0–50 for each compound	At 1 mg/kg: PFOA: 1900 PFOS: 270	Stahl et al. (2009)
Wheat straw	PFAA mixture	Soil + biosolids at different rates: 0.0414–0.220, as ΣPFASs. At the highest rate: PFBA: 0.0135 PFBS: 0.0312 PFOA: 0.0261 PFOS: 0.0408	At 0.220 mg/kg as total amount: PFBA: 22.2 PFBS: 21.8 PFOA: 22.1 PFOS: 11.0 ΣPFASs = 178	Wen et al. (2014)
Wheat grains	PFOA, PFOS	Spiked soil. 0–50 for each compound	At 1 mg/kg: PFOA: 9 PFOS: 0	Stahl et al. (2009)
Wheat grains	PFAA mixture	Soil + biosolids at different rates: 0.0414–0.220, as ΣPFASs. At the highest rate: PFBA: 0.0135 PFBS: 0.0312 PFOA: 0.0261 PFOS: 0.0408	At 0.220 mg/kg as total amount: PFBA: 6.49 PFBS: < MDL PFOA: 2.90 PFOS: 2.53 ΣPFASs = 35.6	Wen et al. (2014)

(continued on next page)

Table 2 (*continued*)

Plant species and organ	Compounds	Type of treatment and initial soil concentrations (mg/kg soil)	Concentrations in plants (μg/kg d.w.)	References
Wheat husks	PFAA mixture	Soil + biosolids at different rates: 0.0414–0.220, as ΣPFASs. At the highest rate: PFBA: 0.0135 PFBS: 0.0312 PFOA: 0.0261 PFOS: 0.0408	At 0.220 mg/kg as total amount: PFBA: 5.77 PFBS: < MDL PFOA: 4.19 PFOS: 2.20 ΣPFASs = 37.8	Wen et al. (2014)

MDL = method-detection limit.
Σ = total amount.

concentrations of PFCAs and PFSAs in straw were those of the C4 perfluorinated chemicals PFBA and PFBS (35,233 ± 6535 and 1843 ± 687 μg/kg d.w., respectively), whereas the concentrations of PFOA and PFOS (both C8 compounds) were lower and similar to each other: 645 ± 121 and 617 ± 178 μg/kg d.w., respectively (Table 2). High amounts of perfluoropentanoic (PFPeA), i.e., 8335 ± 1499 μg/kg d.w., were also recorded. In kernels, the highest PFAS concentrations were those of the C4 - C6 PFCAs, especially PFPeA, with a mean value of 275 ± 91 μg/kg d.w., whereas PFOA content was only 2.45 ± 1.54 μg/kg d.w., and PFBS was the only PFSA detected (5.44 ± 1.90 μg/kg d.w.). Variability in the data, due to the complexity of the extraction procedures and analytical methods utilized, is also evident.

In a full-scale field study using urban biosolids, Blaine et al. (2013) found that PFBA and PFPeA were present only in maize stover, while the other PFASs were below the limit of quantification (LOQ) (Table 2). These results indicate that risk assessment studies based exclusively on PFOA and PFOS accumulation in plants do not give a real picture of the potential dangers to humans of food contamination, as many different PFASs may be present in soil and waters, and consequently in plants.

Comparison of the results reported by Stahl et al. (2009) and Krippner et al. (2015) reveals differences in the extent of PFOA and PFOS accumulation in maize plants, which are particularly evident in the straw (Table 2). Krippner et al. (2015) suggested that these differences could be due to different maize cultivars, growth conditions, and soil characteristics. They also hypothesized that interactions among PFAAs may influence their sorption to soil, and ultimately their uptake and distribution in maize plants.

Hydroponic studies on the uptake of perfluorinated chemicals by cereal plants are less realistic than those using soil as the growth medium, although they can provide more easily interpretable results regarding the mobility of PFASs through the plants without the interference of soil components. Krippner et al. (2014) investigated the accumulation of ten PFCAs and PFSAs with chain lengths from C4 to C10 in maize seedlings grown in nutrient solution and found a U-shaped pattern: five days into the experiment accumulations in the whole plants of the PFCAs with chain lengths up to the C7 compound perfluoroheptanoic acid (PFHpA) decreased, those with chain lengths C7 to C10 increased. Similar results obtained by Müller et al. (2016) with *Arabidopsis thaliana* were attributed to different mechanisms of interaction between plants and PFAS, the long-chained compounds being mainly adsorbed to the root surfaces, the short-chained ones more easily absorbed and translocated. Hydroponic experiments can also reveal information on the partitioning of different PFASs in the various plant compartments, particularly the shoot-to-root concentration ratio, i.e. the translocation factor (TF), which cannot be easily calculated in soil experiments. In maize plants, TF regularly decreased with chain length (< 1 for PFOA and PFCAs with chain lengths > C8), indicating that these chemicals are mainly retained by the root, as also reported by Wen et al. (2013) for PFOA and PFOS. PFSAs also had a similar, or even more pronounced, dependence of retention in roots with increasing chain length, PFBS being the only compound with a high TF. In summary, maize seedlings tend to accumulate C4 - C7 PFCAs and PFBS in the shoots, while the PFAS with higher chain lengths tend to be restricted to the roots. Interestingly, unlike experiments with spiked soil, with hydroponics PFOS accumulated to a greater extent than PFOA, as Wen et al. (2013) also showed, highlighting the important role played by soil components in determining PFAS uptake by plants.

Further results on PFAS accumulation in food crops were obtained by Stahl et al. (2013), who investigated the accumulation of PFOA and PFOS in straw and grain in a five-year lysimeter experiment with the crop sequence winter wheat, winter rye, canola, winter wheat, and winter barley, and with a spiking dose of 25 mg/kg soil. Results confirm that more of these perfluorinated compounds were allocated in the straw than in the grains of all the species examined, in agreement with other reports (Krippner et al., 2015; Stahl et al., 2009). With the exception of canola, PFOS concentrations in the straw were similar to or higher than those of PFOA, with the highest levels measured in the first year of the experiment in winter wheat (14,500 μg/kg d.w. of PFOS vs. 2030 μg/kg d.w. of PFOA). The unusual preponderance of PFOS over PFOA observed in this study was explained by high PFOA losses by leaching. PFBA and PFBS were also detected, these being by-products of the technical mixtures of PFOA and PFOS that were applied to the soil.

A field study on PFAS accumulation in wheat plants grown in biosolid-amended soils confirmed sewage sludge to be an important source of perfluorinated chemicals, and found their concentrations in wheat plants to increase with increasing amendment rate, although in a less-than-proportional manner (Wen et al., 2014). Soil organic matter was suggested as an important factor limiting the uptake of PFASs by wheat roots. Regarding the distribution within the plant of the nine C4–C14 PFCAs and three PFSAs studied, results confirmed the order of accumulation rate straw > grain reported by Krippner et al. (2015) and Stahl et al. (2009, 2013), and the more general order root > straw > grain ≥ husk. The presence of PFAS in husks in concentrations similar to those in grains raises concerns about the possible intake of PFAS with whole wheat flour, which is considered an important source of dietary fiber. PFAS chain length was also a determining factor in plant translocation: the longer the C-chain, the lower the transfer to wheat straw and grains, as shown in maize plants (Krippner et al., 2014, 2015).

Once again, comparison of results from different experiments reveals important differences. For example, Wen et al. (2014) found that at the lowest biosolid dose PFOA and PFOS in wheat grains were at levels of 1.01 ± 0.10 and 0.80 ± 0.02 μg/kg d.w., respectively, starting from soil concentrations of 4.33 ± 0.45 and 10.4 ± 1.0 μg/kg, respectively. Conversely, Stahl et al. (2009) found PFOS in grains only starting from a spiking dose of 10 mg/kg soil, with the contents of PFOA and PFOS in grains at 1300 and 2 μg/kg d.w., respectively.

A more comprehensive parameter with which to compare different experiments and plant species with respect to the plant's ability to accumulate pollutants could be the bioconcentration factor (BCF), i.e. the plant tissue-to-soil concentration ratio (Table 3), for which values > 1 indicate the tendency of plants to accumulate PFAS against a concentration gradient. The C4 compound PFBA had the highest BCF among all the PFASs in both maize and wheat straw. In maize plants grown at spiking doses of 0.25 and 1 mg/kg soil, the BCF for this compound was significantly higher at the lowest pollutant levels i.e. 63.64 *vs* 35.23 (Table 3, Fig. 1) (Krippner et al., 2015). This may indicate some kind of adaptation by plants to high PFBA concentrations or a sort of competition/interaction among PFAS molecules, as suggested by the same authors, although further studies are needed to support these hypotheses. A similar BCF of 64.8 was calculated in the stover of maize plants grown in soil amended with urban biosolids containing 0.10 μg/kg of PFBA (Blaine et al., 2013). In wheat straw, the BCFs for PFBA were more than an order of magnitude lower, ranging

Table 3
Crop plant part analyzed, initial PFAS concentrations (mg/kg soil), soil organic matter (SOM) or organic carbon (OC) contents and bioconcentration factors (concentration in plant parts/initial concentration in soil).

Plant species and parts	Compounds and initial concentrations (mg/kg)	PFBA, PFBS bioconcentration factors	PFOA, PFOS bioconcentration factors	SOM or OC contents	References
Maize straw	PFOA: 0–50 PFOS: 0–50	–	Values at 0.25 and 1 mg/Kg: PFOA: 0.272; 0.126 PFOS: 0.132; 0.104	na	Stahl et al. (2009)
Maize straw	PFAA mixture: 0.25 and 1 per each of 10 compounds	Values at 0.25 and 1 mg/Kg: PFBA: 63.64; 35.23 PFBS: 3.85[a]; 1.84[a]	Values at 0.25 and 1 mg/Kg: PFOA: 0.56[a]; 0.65[a] PFOS: 0.32; 0.62	OC: 0.274%	Krippner et al. (2015)
Maize stover	Full-scale field study, PFAA mixture from urban 2x biosolid addition. PFBA: 0.00010 PFBS: 0.00039 PFOA: 0.00128 PFOS: 0.00282	PFBA: 64.8 PFBS: –	PFOA: – PFOS: –	OC: 0.57%	Blaine et al. (2013)
Maize leaves	PFOS: 38.5 PFBS: 0.02	PFBS: 4.00[b]	PFOS: 0.80[b]	na	Navarro et al. (2017)
Maize ears	PFOA: 0–50 PFOS: 0–50	–; –	Values at 0.25 and 1 mg/Kg: PFOA: 0.008; 0.004 PFOS: 0; 0.003	na	Stahl et al. (2009)
Maize grains	PFAA mixture: 0.25 and 1 per each of 10 compounds	Values at 0.25 and 1 mg/Kg: PFBA: 0.133[a]; 0.229[a] PFBS: 0.008[a]; 0.005[a]	Values at 0.25 and 1 mg/Kg: PFOA: –; 0.002 PFOS: –; –	OC: 0.274%	Krippner et al. (2015)
Maize root	PFOS: 38.5 PFBS: 0.02	PFBS: 5.00[b]	PFOS: 8.82[b]	na	Navarro et al. (2017)
Oat straw	PFOA: 0–50 PFOS: 0–50	–	Values at 0.25 and 1 mg/Kg: PFOA: 0.88; 0.69 PFOS: 0.224; 0.150	na	Stahl et al. (2009)
Oat grains	PFOA: 0–50 PFOS: 0–50	–	Values at 0.25 and 1 mg/Kg: PFOA: 0.048; 0.054 PFOS: 0.004; 0.0170	na	Stahl et al. (2009)
Ryegrass, four cuttings	PFOA: 0–50 PFOS: 0–50	–	Range within the four cuttings at 0.25 and 1 mg/Kg: PFOA: 0.128–7.52 PFOS: 0.048–0.47	na	Stahl et al. (2009)
Wheat straw	PFOA: 0–50 PFOS: 0–50		Values at 0.25 and 1 mg/Kg: PFOA: 3.20; 1.90 PFOS: 0.20; 0.27	na	Stahl et al. (2009)
Wheat straw	PFAA mixture from the highest biosolid addition: PFBA: 0.0135 PFBS: 0.0312 PFOA: 0.0261 PFOS: 0.0408	PFBA: 1.64 PFBS: 0.635	PFOA: 0.847 PFOS: 0.270	SOM: 2.76%	Wen et al. (2014)
Wheat grains	PFOA: 0–50 PFOS: 0–50	–	Values at 0.25 and 1 mg/Kg: PFOA: 0.096; 0.009 PFOS: –; –	na	Stahl et al. (2009)
Wheat grains	PFAA mixture from the highest biosolid addition: PFBA: 0.0135 PFBS: 0.0312 PFOA: 0.0261 PFOS: 0.0408	PFBA: 0.48 PFBS: –	PFOA: 0.111 PFOS: 0.062	SOM: 2.76%	Wen et al. (2014)
Wheat husk	PFAA mixture from the highest biosolid addition: PFBA: 0.0135 PFBS: 0.0312 PFOA: 0.0261 PFOS: 0.0408	PFBA: 0.43 PFBS: –	PFOA: 0.160 PFOS: 0.054	SOM: 2.76%	Wen et al. (2014)

na = not available.
Σ = total amount.
[a] No significant difference between the two different concentrations.
[b] Obtained from mean values of t= 0 and t = final soil concentrations.

from 2.56 to 1.64 with increasing rates of biosolid amendment (Wen et al., 2014). The opposite pattern was observed in grains, where the BCFs for PFBA were higher in wheat, ranging from 0.916 to 0.481 with increasing rates of biosolid amendment (Wen et al., 2014), but were on average 0.18 ± 0.07 in maize grown at spiking doses of 0.25 and 1 mg/kg (Krippner et al., 2015) (Fig. 1). PFPeA was also detected in maize grains at both soil concentrations, but was detected in wheat only at the highest biosolid dose. Recently, Liu et al. (2017) reported very high BCFs for PFBA in maize and wheat grains (on average 2.5 and 33.1, respectively) from an area around a mega-fluorochemical industrial park, suggesting that the contribution of PFASs derived from precipitation as well as from irrigation water. According to these authors, the higher BCFs observed in wheat grains compared to maize might be related to their higher protein content, in accordance with Wen et al.'s (2016) results.

Regarding PFBS in straw, the BCFs of maize, i.e. 3.85 and 1.84 at

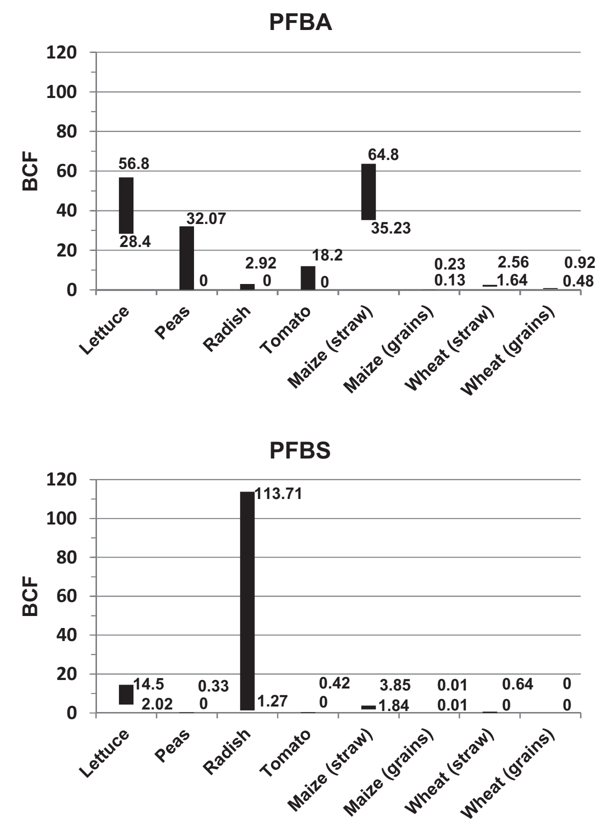


Fig. 1. Bioconcentration factors (BCF) for PFBA and PFBS in vegetable edible plant parts and in maize and wheat straw and grains. Data from Blaine et al., (2013, 2014a) for vegetables, from Krippner et al. (2015) and Blaine et al. (2013) for maize, from Wen et al. (2014) for wheat.

0.25 and 1 mg/kg spiking doses, respectively, (Krippner et al., 2015) were higher than those of wheat, for which the highest value registered was 0.635 (Wen et al., 2014). In grains, PFBS had very low BCFs of 0.008 and 0.005 in maize, and was not detected in wheat by Wen et al. (2014), although Stahl et al. (2013) found PFBS in wheat grains in their lysimeter experiment (448 µg/kg d.w in the first year).

Concerning the C8 PFASs, from Stahl et al.'s (2009) data on maize straw we were able to calculate BCFs of 0.272 and 0.126 for PFOA, and 0.132 and 0.104 for PFOS at 0.25 and 1 mg/kg, respectively. Higher BCFs were reported by Krippner et al. (2015), on average 0.60 ± 0.06 for PFOA, and 0.32 and 0.62 for PFOS at the same 0.25 and 1 mg/kg spiking doses, respectively. BCF values for PFOA in wheat straw were higher than in maize, ranging from 1.54 to 0.847 in Wen et al.'s (2014) experiments, while Stahl et al. (2009) obtained values of 3.20 and 1.90 at the 0.25 and 1 mg/kg doses, respectively, in their trials with wheat. These two authors reported more similar BCFs for PFOS, ranging between 0.20 and 0.27. The BCFs for PFOA and PFOS in maize grains were quite low, ranging from 0 to 0.008 in Stahl et al.'s (2009) and Krippner et al.'s (2015) experiments at the 0.25 and 1 mg/kg spiking doses. The BCFs for PFOA in wheat seeds calculated by Wen et al. (2014) were higher, varying between 0.233 and 0.111, whereas in Stahl et al.'s (2009) experiments they were 0.096 and 0.0090 at the 0.25 and 1 mg/kg spiking doses. BCF values for PFOS obtained from Wen et al.'s (2014) trials were roughly constant with a mean value of 0.0654, whereas from Stahl et al.'s (2009) data they were 0 at both the 0.25 and 1 mg/kg spiking doses.

In summary, overall results regarding the two most important cereal species, maize and wheat, showed a certain variability in BCFs within the same species when comparisons were made between different authors, and between different PFAS spiking concentrations reported by the same authors (see ranges in Fig. 1 for PFBA and PFBS). This confirms the importance of individuating and assessing the parameters that influence PFAS accumulation in plants. Anyway, despite the small number of papers dealing with this topic, it appears that maize tends to accumulate high levels of low-chained fluorinated compounds, particularly PFBA but also PFPeA, in the green parts of the plants to a far greater extent than wheat does. Concerning the C8 compounds, wheat straw seems to accumulate more PFOA than maize straw. Therefore, leaving maize or wheat residues in the soil as amendment in a polluted area may pose serious problems of PFAS spread into the environment, while using them in animal feeds can contaminate the food chain. Conversely, wheat grains and husks, and their derivatives might be a greater direct source of PFBA for humans than maize. However, as there are few papers dealing with these cereals, it is necessary to continue research in order to obtain more data with which additional, more comprehensive analyses can be carried out.

4.2. *Vegetables and fruits*

Stahl et al. (2009) produced the first report on PFAS accumulation not just in cereals but also in vegetable plants. They analyzed PFOA and PFOS concentrations in potato peel and peeled potato tubers from plants grown in soil spiked with the two chemicals in increasing doses from 0 to 50 mg/kg soil. At 1.0 mg/kg of soil, neither PFOA nor PFOS were detectable in peeled potato tubers, whereas in the fibrous structure (peel) they were found in quantities of 2 and 13 µg/kg fresh weight (f.w.), respectively. Studies have also been made of PFOA and PFOS accumulation in carrot, potato and cucumber plants grown in soil treated with contaminated sewage sludge, and in some cases spiked with the two contaminants (Lechner and Knapp, 2011). This research shows that shoot is again an important site of accumulation, whereas the edible parts of these vegetables were, fortunately, less contaminated, following the hierarchy peeled potatoes < < cucumber < peeled carrots. Carrots were the most contaminated of these three vegetable species (Table 4), probably because of their role in absorbing nutrients and water for the whole plant, whereas potatoes are storage organs and cucumber are fruits. Carrot peel contained roughly the same level of PFASs as peeled carrots. Conversely, in the case of potatoes, the peel contained about 2.3 times the concentration of PFOA and 21 times the concentration of PFOS as the peeled tubers, i.e. 17.6 ± 1.4 and 15 ± 2.2 µg/kg f.w., respectively, at the highest spiking dose. Concentrations similar to those found by Lechner and Knapp (2011) were reported by Cornelis et al. (2012) for potatoes sampled in Belgium. However, the PERFOOD study conducted in four European countries with the purpose of studying the risk to human health due to the presence of PFASs in European foods, found that the sum of PFHxA, PFHpA, PFOA and perfluorononanoic acid (PFNA) levels in starchy roots and tubers were on the scale of tens of nanograms per kg of f.w. (Herzke et al., 2013), i.e. approximately two orders of magnitude lower than the PFOA concentrations found by Lechner and Knapp (2011). Bizkarguenaga et al. (2016) used LUFA 2.4 soil (LUFA Speyer, Speyer, Germany) mixed with compost spiked with PFOA and PFOS at levels similar to those used by Lechner and Knapp (2011), and found comparable amounts of these two compounds in carrot peel and cores, with PFOS in generally higher quantities than PFOA. The authors also reported lower plant accumulation of the two PFASs when a commercial universal substrate was used, which they attributed to its higher organic matter content (53% vs 2.3% of total organic carbon). Large differences in the PFAS concentrations between the two carrot varieties studied were found, but only with the commercial universal substrate for PFOA (121 ± 36 µg/kg in *Chantenay vs* 41 ± 11 µg/kg in *Nantesa*), and with the LUFA 2.4 soil for PFOS (240 ± 22 in *Chantenay* vs 162 ± 17 in *Nantesa*), indicating different effects of soil characteristics on PFAS uptake by the two varieties.

Other studies on PFAS accumulation in vegetables were conducted with lettuce leaves and tomato fruits (Blaine et al., 2013), and in different organs of radish, celery, tomato, and sugar snap pea plants

Table 4
Initial PFAS concentrations (mg/kg soil) and their amount in different vegetable plant parts (μg/kg d.w.).

Plant Species and parts	Compounds	Type of treatment and initial soil concentrations	Concentrations in plant tissues	References
Carrots (peeled)	PFOA; PFOS Tub 1	Soil + spiked biosolids. 0.681; 0.010	PFOA: 31.3[a]; 333[b] PFOS: 0.5[a]; 5.3[b]	Lechner and Knapp (2011)
Carrots (peeled)	PFOA; PFOS Tub 2	Soil + spiked biosolids. 0.676; 0.458	PFOA: 30.8[a]; 328[b] PFOS: 18.4[a]; 196[b]	Lechner and Knapp (2011)
Carrots (peeled), *Chantenay* variety	PFOA and PFOS in separate pots	Soil 2.4 + spiked compost. 0.528; 0.445	PFOA: 148 PFOS: 240	Bizkarguenaga et al. (2016)
Carrots (peeled), *Nantesa* variety	PFOA and PFOS in separate pots	Soil 2.4 + spiked compost. 0.485; 0.335	PFOA: 144 PFOS: 162	Bizkarguenaga et al. (2016)
Celery shoots	PFAA mixture	Soil + biosolids. Industrially impacted: PFBA: 0.00468 PFBS: 0.04858 PFOA: 0.07852 PFOS: 0.04966 ΣPFASs = 0.329;	PFBA: 231.69 PFBS: 107.13 PFOA: 55.40 PFOS: 69.27 ΣPFASs = 817.3	Blaine et al. (2014a)
Celery shoots	PFAA mixture	Soil + biosolids. Municipal: PFBA: 0.00090 PFBS: 0.00021 PFOA: 0.01491 PFOS: 0.31949 ΣPFASs = 0.427	PFBA: $<$ LOQ PFBS: 4.49 PFOA: 1.99 PFOS: 17.21 ΣPFASs = 39.3	Blaine et al. (2014a)
Cucumbers Tub 1	PFOA; PFOS	Soil + spiked biosolids. 0.406; 0.010	PFOA: 11.3[a]; 323[c] PFOS: ND	Lechner and Knapp (2011)
Cucumbers Tub 2	PFOA; PFOS	Soil + spiked biosolids. 0.805; 0.556	PFOA: 23.8[a]; 680[c] PFOS: 1.3[a]; 37.1[c]	Lechner and Knapp (2011)
Lettuce leaves	PFAA mixture	Soil + biosolids. Industrially impacted: PFBA: 0.00468 PFBS: 0.04858 PFOA: 0.07852 PFOS: 0.04966 ΣPFASs = 0.329	PFBA: 266.08 PFBS: 205.24 PFOA: 197.91 PFOS: 82.90 ΣPFASs = 1245	Blaine et al. (2013)
Lettuce leaves	PFAA mixture	Soil + biosolids. Municipal: PFBA: 0.00090 PFBS: 0.00021 PFOA: 0.01491 PFOS: 0.31949 ΣPFASs = 0.427	PFBA: 25.50 PFBS: 3.03 PFOA: 20.01 PFOS: 101.62 ΣPFASs = 240	Blaine et al. (2013)
Lettuce leaves	Field-scale trial plots PFAA mixture	Soil + biosolids, 4 × PFBA: 0.00069 PFBS: 0.00080 PFOA: 0.00517 PFOS: 0.01391 ΣPFASs = 0.03477	PFBA: 27.5 PFBS: 1.62 PFOA: $<$ LOQ PFOS: 1.39 ΣPFASs = 47.43	Blaine et al. (2013)
Lettuce leaves	PFOA and PFOS in separate pots	Soil 2.4 + spiked compost: 0.560; 0.508	PFOA: 1029 PFOS: 77	Bizkarguenaga et al. (2016)
Pea fruits	PFAA mixture,	Soil + biosolids. Industrially impacted: PFBA: 0.00468 PFBS: 0.04858 PFOA: 0.07852 PFOS: 0.04966 ΣPFASs = 0.329	PFBA: 150.14 PFBS: 16.18 PFOA: 2.65 PFOS: 1.28 ΣPFASs = 236	Blaine et al. (2014a)
Pea fruits	PFAA mixture	Soil + biosolids. Municipal: PFBA: 0.00090 PFBS: 0.00021 PFOA: 0.01491 PFOS: 0.31949 ΣPFASs = 0.427	PFBA: $<$ LOQ PFBS: $<$ LOQ PFOA: $<$ LOQ PFOS: $<$ LOQ ΣPFASs ≤ LOQ	Blaine et al. (2014a)
Potatoes (peeled)	PFOA, PFOS	Spiked soil. 0–50 for each compound	At 1 mg/Kg soil: PFOA: 0 PFOS: 0	Stahl et al. (2009)
Potatoes (peeled)	PFOA; PFOS Tub 1	Soil + spiked biosolids. 0.276; 0.015	PFOA: 2.9[a]; 17.9[b] PFOS: $<$ LOD	Lechner and Knapp (2011)
Potatoes (peeled)	PFOA; PFOS Tub 2	Soil + spiked biosolids. 0.795; 0.317	PFOA: 7.7[a]; 47.5[b] PFOS: 0.7[a]; 4.3[b]	Lechner et al. (2011)
Radish roots	PFAA mixture	Soil + biosolids. Industrially impacted: PFBA: 0.00468 PFBS: 0.04858 PFOA: 0.07852 PFOS: 0.04966 ΣPFASs = 0.329	PFBA: 13.67 PFBS: 61.89 PFOA: 66.89 PFOS: 34.86 ΣPFASs = 279	Blaine et al. (2014a)

(continued on next page)

Table 4 (*continued*)

Plant Species and parts	Compounds	Type of treatment and initial soil concentrations	Concentrations in plant tissues	References
Radish roots	PFAA mixture	Soil + biosolids. Municipal: PFBA: 0.00090 PFBS: 0.00021 PFOA: 0.01491 PFOS: 0.31949 ΣPFASs = 0.427	PFBA: < LOQ PFBS: 23.88 PFOA: 8.11 PFOS: 21.03 ΣPFASs = 79.3	Blaine et al. (2014a)
Spinach	PFAA mixture	Soil + biosolids W1: PFBA: 0.00045 PFBS: ND PFOA: 0.00019 PFOS: 0.00035 ΣPFASs = 0.00144	PFBA: ND PFBS: ND PFOA: 2.37 PFOS: 1.62 ΣPFASs = 5.33	Navarro et al. (2017)
Spinach	PFAA mixture	Soil + biosolids W2: PFBA: 0.00043 PFBS: ND PFOA: 0.00018 PFOS: 0.00022 ΣPFASs = 0.00096	PFBA: ND PFBS: ND PFOA: ND PFOS: 0.99 ΣPFASs = 0.99	Navarro et al. (2017)
Tomato fruits	PFAA mixture	Soil + biosolids. Industrially impacted: PFBA: 0.00468 PFBS: 0.04858 PFOA: 0.07852 PFOS: 0.04966 ΣPFASs = 0.329	PFBA: 56.11 PFBS: 19.38 PFOA: 8.81 PFOS: < LOQ ΣPFASs = 337	Blaine et al. (2013)
Tomato fruits	PFAA mixture	Soil + biosolids. Municipal: PFBA: 0.00090 PFBS: 0.00021 PFOA: 0.01491 PFOS: 0.31949 ΣPFASs = 0.427	PFBA: < LOQ PFBS: < LOQ PFOA: < LOQ PFOS: < LOQ ΣPFASs = 21.4	Blaine et al. (2013)
Tomato fruits	Field-scale trial plots PFAA mixture	Soil + biosolids, 4× PFBA: 0.00069 PFBS: 0.00080 PFOA: 0.00517 PFOS: 0.01391 ΣPFASs = 0.03477	PFBA: 12.56 PFBS: < LOQ PFOA: < LOQ PFOS: < LOQ ΣPFASs = 37.7	Blaine et al. (2013)
Tomato fruits	PFAA mixture	Soil + biosolids W1: PFBA: 0.00074 PFBS: ND PFOA: 0.0012 PFOS: 0.00047 Σ = 0.00782	PFBA: 12.45 PFBS: ND PFOA: 0.18 PFOS: 0.03 ΣPFASs = 61.30	Navarro et al. (2017)
Tomato fruits	PFAA mixture	Soil + biosolids W2: PFBA: 0.00007 PFBS: 0.00013 PFOA: 0.00024 PFOS: 0.00030 ΣPFASs = 0.00088	PFBA: 2.48 PFBS: ND PFOA: ND PFOS: ND ΣPFASs = 3.47	Navarro et al. (2017)

LOD = limit of detection.
LOQ = limit of quantitation.
ND = not detected.
Σ = total amount.
[a] On a fresh weight basis.
[b] On a dry weight basis calculated assuming the % dry matter values of 9.4 and 16.2 (author's data) for peeled carrots and potatoes, respectively.
[c] Calculated on a dry weight basis assuming a 3.5% dry matter content (http://nut.entecra.it/646/tabelle_di_composizione_degli_alimenti.html).

(*Pisum sativum* var. *macrocarpon*) grown in soil amended with municipal biosolids or industrially impacted municipal biosolids (Blaine et al., 2014a). Examination of the concentrations of different PFAAs, consisting of C4 to C10 PFCAs and PFSAs, revealed much higher levels of contamination in plants grown in soil amended with biosolids, particularly those containing industrial residues (Table 4), compared with control soil. For example, PFOS concentrations were < LOQ in lettuce grown in control soil, 82.90 ± 15.93 µg/kg in soil amended with municipal biosolids, and 101.62 ± 5.80 µg/kg in soil amended with industrially impacted municipal biosolids, confirming sludges as potential sources of PFASs in plants. Similar conclusions were reached by Navarro et al. (2017), who detected PFASs in anaerobically digested thermal drying sludge (W1) and in anaerobically digested municipal solid waste compost (W2). They found high rates of PFBA and PFPeA accumulation in fruits of tomato plants grown in amended soil, especially with W1 (Table 4). Analyses carried out by Blaine et al., (2013, 2014a) of the bioconcentration factors for plants grown in soil amended with industrially impacted municipal biosolids revealed that PFBA had the highest values in almost all the organs of the lettuce, radish, celery and pea plants examined, increasing from the roots to the aboveground part of the plants, and reaching the highest levels in the edible parts of the lettuce (56.8 ± 3.45), celery (49.49 ± 4.50) and pea (32.07 ± 32.07) plants, and in tomato shoot (26.03 ± 2.28). However, tomato fruits had a slightly higher BCF for PFPeA than for PFBA (17.06 ± 3.74 *vs* 12.16 ± 1.71). Conversely, in the aboveground parts of these plants, PFOA and PFOS tended to have somewhat lower BCF values (Table 5). BCFs for plants grown in soil amended with industrially impacted municipal biosolids were generally higher than

Table 5
Vegetable plant part analyzed, initial PFAS concentrations (mg/kg soil), soil or sludge organic carbon (OC) contents and bioconcentration factors (concentration in plant parts/initial concentration in soil).

Plant species and parts	Compounds and initial concentrations (mg/kg)	PFBA, PFBS bioconcentration factors	PFOA, PFOS bioconcentration factors	Soil or sludge OC contents	References
Carrots (peeled)	PFOA: 0.681 PFOS: 0.010	–	PFOA: 0.49[a] PFOS: 0.53[a]	na	Lechner et al. (2011)
Carrots (peeled)	PFOA: 0.676 PFOS: 0.458	–	PFOA: 0.49[a] PFOS: 0.43[a]	na	Lechner et al. (2011)
Carrots (peeled), *Chantenay* variety	PFOA: 0.528 PFOS: 0.445 in separate pots	–	PFOA: 0.28 PFOS: 0.55	Soil OC: 4.9%[b]	Bizkarguenaga et al. (2016)
Carrots (peeled), *Nantesa* variety	PFOA: 0.485; PFOS: 0.335 in separate pots	–	PFOA: 0.30 PFOS: 0.49	Soil OC: 4.9%[b]	Bizkarguenaga et al. (2016)
Celery shoots	PFAA mixture, from Industrially impacted biosolids: PFBA: 0.00468 PFBS: 0.04858 PFOA: 0.07852 PFOS: 0.04966 ΣPFASS = 0.329	PFBA: 49.49 PFBS: 2.21	PFOA: 0.71 PFOS: 1.39	Soil OC: 2.24%	Blaine et al. (2014a)
Celery shoots	PFAA mixture from Municipal biosolids: PFBA: 0.00090 PFBS: 0.00021 PFOA: 0.01491 PFOS: 0.31949 ΣPFASS = 0.427	PFBA: – PFBS: 21.4	PFOA: 0.13 PFOS: 0.05	Soil OC: 6.34%	Blaine et al. (2014a)
Cucumbers Tub 1	PFOA: 0.406 PFOS: 0.010	–	PFOA: 0.79[b] PFOS: –	na	Lechner et al. (2011)
Cucumbers Tub 2	PFOA: 0.805 PFOS: 0.556	–	PFOA: 0.85[b] PFOS: 0.067[b]	na	Lechner et al. (2011)
Lettuce leaves	PFAA mixture from Industrially impacted biosolids: PFBA: 0.00468 PFBS: 0.04858 PFOA: 0.07852 PFOS: 0.04966 ΣPFASS = 0.329	PFBA: 56.8 PFBS: 4.22	PFOA: 2.52 PFOS: 1.67	Soil OC: 2.24%	Blaine et al. (2013)
Lettuce leaves	PFAA mixture from Municipal biosolids: PFBA: 0.00090 PFBS: 0.00021 PFOA: 0.01491 PFOS: 0.31949 ΣPFASS = 0.427	PFBA: 28.4 PFBS: 14.5	PFOA: 1.34 PFOS: 0.32	Soil OC: 6.34%	Blaine et al. (2013)
Lettuce leaves	Field-scale trial plots, PFAA mixture from biosolid addition 4×: PFBA: 0.00069 PFBS: 0.00080 PFOA: 0.00517 PFOS: 0.01391 ΣPFASS = 0.03477	PFBA: 40.0 PFBS: 2.02	PFOA: – PFOS: 0.10	Soil OC: 3.51%	Blaine et al. (2013)
Lettuce leaves	PFOA: 0.560 PFOS: 0.507 in separate pots	–	PFOA: 1.85 PFOS: 0.15	Soil OC: 4.9% [b]	Bizkarguenaga et al. (2016)
Pea fruits	PFAA mixture from Industrially impacted biosolids: PFBA: 0.00468 PFBS: 0.04858 PFOA: 0.07852 PFOS: 0.04966 ΣPFASS = 0.329	PFBA: 32.07 PFBS: 0.33	PFOA: 0.03 PFOS: 0.03	Soil OC: 2.24%	Blaine et al. (2014a)
Pea fruits	PFAA mixture from Municipal biosolids: PFBA: 0.00090 PFBS: 0.00021 PFOA: 0.01491 PFOS: 0.31949 ΣPFASS = 0.427	PFBA: – PFBS: –	PFOA: – PFOS: –	Soil OC: 6.34%	Blaine et al. (2014a)
Potatoes (peeled)	PFOA: 0 – 50 PFOS: 0 – 50	–	Values at 0.25 and 1 mg/Kg: PFOA: 0; 0 PFOS: 0; 0	na	Stahl et al. (2009)
Potatoes (peeled) Tub 1	PFOA: 0.276 PFOS: 0.015	–	PFOA: 0.065[b] PFOS: –	na	Lechner et al. (2011)
Potatoes (peeled) Tub 2	PFOA: 0.795 PFOS: 0.317	–	PFOA: 0.045[b] PFOS: 0.010[b]	na	Lechner et al. (2011)

(*continued on next page*)

Table 5 (*continued*)

Plant species and parts	Compounds and initial concentrations (mg/kg)	PFBA, PFBS bioconcentration factors	PFOA, PFOS bioconcentration factors	Soil or sludge OC contents	References
Radish roots	PFAA mixture from Industrially impacted biosolids: PFBA: 0.00468 PFBS: 0.04858 PFOA: 0.07852 PFOS: 0.04966 ΣPFASS = 0.329	PFBA: 2.92 PFBS: 1.27	PFOA: 0.85 PFOS: 0.70	Soil OC: 2.24%	Blaine et al. (2014a)
Radish roots	PFAA mixture from Municipal biosolids: PFBA: 0.00090 PFBS: 0.00021 PFOA: 0.01491 PFOS: 0.31949 ΣPFASS = 0.427	PFBA: – PFBS: 114	PFOA: 0.54 PFOS: 0.066	Soil OC: 6.34%	Blaine et al. (2014a)
Tomato fruits	PFAA mixture from Industrially impacted biosolids: PFBA: 0.00468 PFBS: 0.04858 PFOA: 0.07852 PFOS: 0.04966 ΣPFASS = 0.329	PFBA: 12.2 PFBS: 0.42	PFOA: 0.11 PFOS: –	Soil OC: 2.24%	Blaine et al. (2013) and Blaine et al. (2014a)
Tomato fruits	PFAA mixture from Municipal biosolids: PFBA: 0.00090 PFBS: 0.00021 PFOA: 0.01491 PFOS: 0.31949 ΣPFASS = 0.427	PFBA: – PFBS: –	PFOA: – PFOS: –	Soil OC: 6.34%	Blaine et al. (2013) and Blaine et al. (2014a)
Tomato fruits	Field-scale trial plots, PFAA mixture from biosolid addition 4×: PFBA: 0.00069 PFBS: 0.00080 PFOA: 0.00517 PFOS: 0.01391 ΣPFASS = 0.03477	PFBA: 18.2 PFBS: –	PFOA: – PFOS: –	Soil OC: 3.51%	Blaine et al. (2013)

na = not available.
[a] Calculated on a dry weight basis from Table 4 data.
[b] Calculated taking into account the % OC content of both soil and spiked compost, and their mixture ratio.

Table 6
Bioconcentration factors calculated for lettuce and celery grown in soil amended with industrially impacted biosolids or municipal biosolids.

Compound	Soil + industrially impacted biosolids Lettuce leaves	Soil + municipal biosolids Lettuce leaves	Soil + industrially impacted biosolids Celery shoots	Soil + municipal biosolids Celery shoots
PFBA	56.8	28.4	49.5	< LOQ
PFPeA	20.4	10.2	12.8	1.12
PFHxA	9.90	11.7	11.9	3.00
PFHpA	2.66	3.33	2.51	0.66
PFOA	2.52	1.34	0.71	0.13
PFNA	2.85	0.77	0.69	0.27
PFDA	0.52	0.34	0.32	0.20
PFBS	4.22	14.5	2.21	21.4
PFHxS	7.56	1.08	2.31	0.07
PFOS	1.67	0.32	1.39	0.05

Values taken or calculated from soil and plant data of the greenhouse study of Blaine et al. (2013, 2014a).

when the soil was amended with municipal biosolids, with the exception of PFBS, for which celery and lettuce plants treated with municipal biosolids had the higher BCF values (Table 6; Blaine et al., 2013; Blaine et al., 2014a). This clearly shows that the amount of PFASs accumulated from biosolid-treated soils by the same plant species depends on the kind of sludge utilized, not only with respect to the total and individual concentrations of these pollutants, but also because different biosolids have different effects on the availability and/or uptake of individual PFASs by plants. Blaine et al. (2013) hypothesized that differences in the bioavailability of PFAAs may also be due to the different nature of the organic carbon of the two kinds of biosolids.

Hydroponic studies on the uptake of perfluorinated chemicals are probably more realistic when they concern vegetable plants rather than cereals as many vegetables can be grown in this manner. However, calculating BCFs and comparing them with those obtained from plants grown in soil is not easy owing to the changes in nutrient solution volumes, and consequently in PFAS concentrations, due to evapotranspiration. Felizeter et al. (2012, 2014) studied PFAS accumulation in lettuce, tomato, cabbage and zucchini plants grown in a nutrient solution containing 14 PFASs, comprising the C4 to C14 PFCAs, and the C4, C6 and C8 PFSAs. Lettuce tended to accumulate the lighter C4 - C7 chemicals in the leaf, in accordance with Blaine et al. (2013), but PFOA and PFCAs with higher chain lengths tended to accumulate in the roots. Likewise, the three PFSAs considered accumulated mainly in the roots, again in increasing percentages with increasing chain lengths. The higher sorption of longer-chained PFASs to plant organic components, and their reduced ability to cross the Casparian strip have been put forward to explain these results (Felizeter et al., 2012). The same effect of chain length on PFAS distribution across plant organs was also found in tomato, cabbage and zucchini plants, with the longer-chained chemicals having a preference for the roots, the shorter-chained ones for the leaves and fruits or heads, see Fig. 2 for the mass distribution of

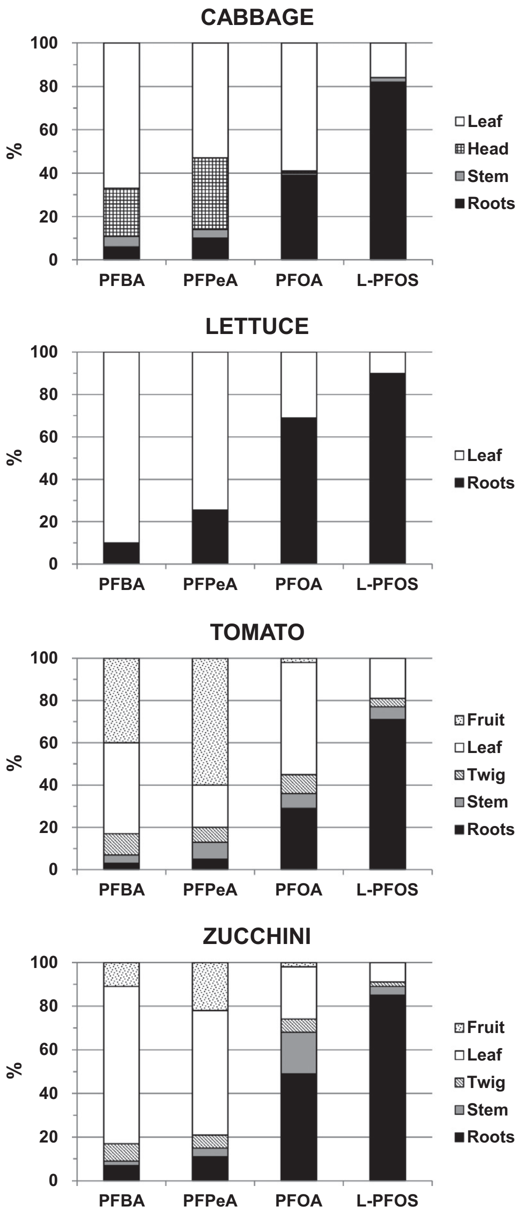


Fig. 2. Mass distribution of some PFAAs in different tissues of cabbage, lettuce, tomato and zucchini plants, expressed as percent of the total amount of PFAA found in the plant. Data from Felizeter et al. (2012, 2014).

some PFAAs in different tissues of cabbage, lettuce, tomato and zucchini plants (Felizeter et al., 2012, 2014). However, it is important to point out that PFAS distribution across roots, stem, leaf, fruit and head varied among species, as seen in Blaine et al.'s (2013; 2013 a) experiments and in cereals. PFOA and PFOS spiked at a nominal concentration of 1 μg/L, with several renewals of the treatment solutions, were accumulated, respectively, to 1.55 ± 0.61 and 0.50 ± 0.31 μg/kg f.w. in lettuce leaves, 0.34 ± 0.10 and 0.10 ± 0.041 μg/kg f.w. in cabbage head, 0.29 ± 0.06 and 0.12 ± 0.023 μg/kg f.w. in zucchini fruit, and 0.28 ± 0.035 μg/kg f.w. and < LOQ in tomato fruit (Felizeter et al., 2012, 2014). It is important to mention that the authors concluded that the quantities of PFOA and PFOS measured in these plants were well below the levels that could pose a risk to human health, as established by the European Food Safety Authority (2008) tolerable daily intake (TDI) values for PFOA and PFOS, i.e. 1500 ng/kg body weight and 150 ng/kg body weight, respectively (EFSA, 2008). Recently, the US EPA specified the lower oral non-cancer reference doses (RfDs) of both PFOS and PFOA at 0.00002 mg/kg/day = 20 ng/kg/day (EPA, 2016a, 2016b). To exceed these values, a person of 70 kg body weight would need to eat about 0.9 kg (for PFOA) and 2.8 kg (for PFOS) of lettuce grown at a concentration of 1 μg/L. At this spiking dose, PFBA accumulated to a greater extent than PFOA and PFOS, that is, to 11.7 ± 2.4 μg/kg f.w. in lettuce, 7.21 ± 3.09 μg/kg f.w. in cabbage, 5.56 ± 1.12 μg/kg f.w. in tomato, and 1.24 ± 0.11 in zucchini. PFPeA accumulated to a similar or greater extent than PFBA in all the vegetables except lettuce. The C4 PFBS accumulated to a lesser extent than PFBA, as is the case with cereals, with a mean concentration of 0.61 ± 0.26 μg/kg f.w. in the vegetable species analyzed (Felizeter et al., 2012, 2014).

In the context of the European project PERFOOD, D'Hollander et al. (2015) found that PFAA detection rates in fruits were generally higher than in other food groups (cereals, sweets, salts), with values depending on species and origin. The most frequently detected PFASs were PFOA followed by PFOS, their concentrations in fruits ranging from < LOQ to 139 and 539 ng/kg f.w., respectively. Dietary intake assessments for PFOA and PFOS made by these authors showed that the daily intake of PFASs was far below the current tolerable levels established by the EFSA. Regarding the short-chain compounds, Klenow et al. (2013) found that fruits accounted for 39%, 58%, 60% and 20% of PFHxA total dietary intake in the Czech Republic, Italy, Norway and Belgium, respectively. The PERFOOD studies contain no data on PFBA and PFPeA, which can accumulate in plants to a greater extent than PFHxA (Table 6).

4.3. Parameters affecting PFAS accumulation in plants

As far as we know, Higgins and Luthy (2006) were the first researchers to point to organic carbon content as a dominant parameter affecting PFAS sorption to sediments. Zhao et al. (2011) showed that soil organic matter, followed by cation exchange capacity (CEC), has the greatest influence on PFOA and PFOS phytotoxicity for *Brassica chinensis* plants. In a soil leaching experiment, Gellrich et al. (2012) found strong adsorption of PFASs to sewage sludge, which they attributed to its high organic carbon content. Zhao et al. (2016) compared PFAA sequestration and bioavailability in farmland soil with two peat soils, and showed that the higher the soil organic matter content, the higher the sequestration of these substances by soils. It must be added that PFAAs were not irreversibly sequestrated, as they were accumulated to some extent by earthworms. Still regarding the role of soil organic matter, Wen et al. (2014) found that BCF values were less variable if they were calculated on the basis of soil organic matter. Likewise, Bizkarguenaga et al. (2016) reported lower plant accumulation of PFOA and PFOS when a substrate rich in organic matter was used. However, Blaine et al. (2014 b) studied the accumulation in lettuce of PFAS from soils differing in their organic carbon contents (0.4%, 2%, 6%), and found greater accumulation of the short-chained compounds PFBA, PFPeA and PFBS at 2% soil organic content than at 0.4 or 6%. Furthermore, as discussed before, Blaine et al. (2014a) suggested that the nature of organic carbon can also influence PFAS bioavailability to plants. Other soil characteristics can affect the amount of PFAS that can be absorbed by plants. Stahl et al. (2013) attributed the unusual lower accumulation of PFOA compared with PFOS to the higher leaching of PFOA from the soil. Zhao et al. (2014) found that the presence of the redworm *Eisenia fetida* in soil induces increased bioaccumulation of PFCAs with $C \leq 7$, and decreased accumulation of those with $C > 7$ and of PFSAs in wheat plants. A further parameter influencing the amount of PFAS accumulated by plants may be their composition. Gellrich et al. (2012) found that PFBA was strongly adsorbed by soil only in the absence of long carbon chain PFAS, probably because of competition for sites of adsorption.

Further environmental parameters that can affect PFAS absorption by plants are temperature and salinity. Zhao et al. (2013, 2016) showed that increases in these parameters induce higher PFCA and PFOS uptake and translocation in wheat plants grown in hydroponics. High temperatures induce higher transpiration rates, which can lead to higher contaminant uptake. High salinity may trigger increased partitioning of PFAS into the organic components of the roots with a salting-out mechanism, as suggested by You et al. (2010) to partially justify the higher sorption of PFOS onto sediments at increasing $CaCl_2$ levels. Regarding the effect of pH, Zhao et al. (2013) found the highest PFOS accumulations in wheat plants grown in nutrient solutions at pH values ranging from 6 to 8, with lower uptake at pH 4 and particularly at pH 10. In contrast, Krippner et al. (2014) did not observe pH dependence for PFOS uptake by maize plants grown in nutrient solution, but found that PFDA accumulated to a greater extent at pH 5 than at pH 7, while the influence of this parameter on other PFCAs was inconsistent. Wen et al. (2015) found soil pH to play a relatively unimportant role in PFOS and PFOA bioavailability to earthworms in biosolid amended soil. Similar views were expressed by Zhao et al. (2011) regarding the role of soil pH and CEC on PFOS and PFOA toxicity for *Brassica chinensis* plants. Additional insights by Blaine et al. (2014 b) suggest that the mobility and bioavailability of PFAAs may be greater when delivered with irrigation water than through biosolid-amended soil. To conclude, Li et al. (2018) recently reviewed available data on the role of soil and sediment properties in determining PFAS sorption and argued that the behavior of these substances in the environment is more complex than can be explained by a single soil or sediment property.

4.4. Challenges for the future and research needs

As discussed previously, PFASs are highly persistent compounds. Even the recently-adopted fluorinated alternatives and their precursors are highly persistent and environmentally mobile, so even if emissions cease in the future, it will take decades or even centuries to reverse the global contamination of short-chain PFAAs (Wang et al., 2015). Owing to the current unfeasibility of removing PFASs from foods, the first challenge to growing safer agricultural products must be to identify the sources of contamination in irrigation water and soil. As already mentioned, waters and soils close to airports, fire-training locations, industrial sites and landfills are particularly at risk of PFAS pollution. However, the use of biosolids on agricultural soil must also be considered a potential source of agricultural plant contamination. In polluted areas, to reduce dietary intake of PFAS, it is important to adopt agricultural measures that decrease their absorption by plants, in particular the addition of unpolluted manure to the soil, as many studies provide evidence for a decrease in PFAS accumulation in plants at increasing levels of soil organic matter. Furthermore, the use of plant residues, such as maize stover, in animal feeds should be avoided as this practice can contaminate the food chain. Growing vegetables in hydroponics should also be discouraged in polluted areas, as it allows pollutants to interact directly with plants in the absence of soil and its ability to adsorb pollutants. The adoption of phytoremediation techniques may counter the spread of PFASs in the environment, as it has been shown that tree plants and constructed wetlands are able to remove these pollutants (Gobelius et al. (2017); Yin et al. (2017). However, as discussed previously, more studies are needed, as many research results display discrepancies regarding the tendency of the same plant species to accumulate PFASs, and the role of soil components in PFAS bioavailability. There is also the need to fill the gaps in the knowledge of the accumulation capacity of PFAS by other plant species. The role of volatile PFAS precursors in the accumulation of these substances by plants also needs to be clarified. The general variability observed in the examined data further complicates our understanding of how PFASs interact with soils and plants.

5. Conclusions

Current research suggests that the spread of PFASs in agricultural soil is mainly a result of irrigation with contaminated water, the use of polluted sewage sludges or industrial wastes as soil conditioners. Agricultural soils especially at risk may be those close to airports and fire-training locations. PFASs are absorbed by plants to different extents according to their concentrations, chain lengths, functional group, plant species and variety, growth media (hydroponics vs. soil), and soil and biosolid characteristics. In particular, the abundance and characteristics of soil organic matter are considered one of the most important factors. Once inside the plants, partitioning among organs depends on species, and, particularly, on functional group and chain length. The C4–C6 compounds, which have recently replaced C8 PFOA and PFOS in many industrial processes, appear to accumulate particularly in leaves and fruits, whereas the compounds with higher chain lengths tend to be more concentrated in roots. However, owing to the variability of available data, the complexity of the interactions between PFASs and soil components, and between PFASs and plants, further studies need to be undertaken in order to clarify the role of each factor influencing the uptake and distribution of these compounds within plants in order to reduce human exposure to these hazardous substances.

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References

Abbott, B.D., Wolf, C.J., Schmid, J.E., Das, K.P., Zehr, R.D., Helfant, L., Nakayama, S., Lindstrom, A.B., Strynar, M.J., Lau, C., 2007. Perfluorooctanoic acid–induced developmental toxicity in the mouse is dependent on expression of peroxisome proliferator–activated receptor-alpha. Toxicol. Sci. 98, 571–581.
Ahrens, L., 2011. Polyfluoroalkyl compounds in the aquatic environment: a review of their occurrence and fate. J. Environ. Monit. 13, 20–31.
Ahrens, L., Barber, J.L., Xie, Z., Ebinghaus, R., 2009. Longitudinal and latitudinal distribution of perfluoroalkyl compounds in the surface water of the Atlantic Ocean. Environ. Sci. Technol. 43, 3122–3127.
Ahrens, L., Shoeib, M., Harner, T., Lee, S.C., Guo, R., Reiner, E.J., 2011a. Wastewater treatment plant and landfills as sources of polyfluoroalkyl compounds to the atmosphere. Environ. Sci. Technol. 45, 8098–8105.
Ahrens, L., Xie, Z., Ebinghaus, R., 2010. Distribution of perfluoroalkyl compounds in seawater from Northern Europe, Atlantic Ocean, and Southern Ocean. Chemosphere 78, 1011–1016.
Ahrens, L., Yeung, L.W.Y., Taniyasu, S., Lam, P.K.S., Yamashita, N., 2011b. Partitioning of perfluorooctanoate (PFOA), perfluorooctane sulfonate (PFOS) and perfluorooctane sulfonamide (PFOSA) between water and sediment. Chemosphere 85, 731–737.
Ankley, G.T., Kuehl, D.W., Kahl, M.D., Jensen, K.M., Linnum, A., Leino, R.L., Villeneuve, D.A., 2005. Reproductive and developmental toxicity and bioconcentration of perfluorooctanesulfonate in a partial life-cycle test with the fathead minnow (*Pimephales promelas*). Environ. Toxicol. Chem. 24, 2316–2324.
Backe, W.J., Day, T.C., Field, J.A., 2013. Zwitterionic, cationic, and anionic fluorinated chemicals in aqueous film forming foam formulations and groundwater from US military bases by nonaqueous large-volume injection HPLC-MS/MS. Environ. Sci. Technol. 47, 5226–5234.
Bischel, H.N., MacManus-Spencer, L.A., Luthy, R.G., 2010. Noncovalent interactions of long-chain perfluoroalkyl acids with serum albumin. Environ. Sci. Technol. 44, 5263–5269.
Bizkarguenaga, E., Zabaleta, I., Prieto, A., Fernández, L., Zuloaga, O., 2016. Uptake of 8: 2 perfluoroalkyl phosphate diester and its degradation products by carrot and lettuce from compost-amended soil. Chemosphere 152, 309–317.
Bjermo, H., Darnerud, P.O., Pearson, M., Barbieri, H.E., Lindroos, A.K., Nälsén, C., Lindh, C.H., Jönsson, B.A., Glynn, A., 2013. Serum concentrations of perfluorinated alkyl acids and their associations with diet and personal characteristics among Swedish adults. Mol. Nutr. Food Res. 57, 2206–2215.
Blaine, A.C., Rich, C.D., Hundal, L.S., Lau, C., Mills, M.A., Harris, K.M., Higgins, C.P., 2013. Uptake of perfluoroalkyl acids into edible crops via land applied biosolids: field and greenhouse studies. Environ. Sci. Technol. 47, 14062–14069.
Blaine, A.C., Rich, C.D., Sedlacko, E.M., Hundal, L.S., Kumar, K., Lau, C., Mills, M.A., Harris, K.M., Higgins, C.P., 2014a. Perfluoroalkyl acid distribution in various plant compartments of edible crops grown in biosolids-amended soils. Environ. Sci. Technol. 48, 7858–7865.
Blaine, A.C., Rich, C.D., Sedlacko, E.M., Hyland, K.C., Stushnoff, C., Dickenson, E.R., Higgins, C.P., 2014b. Perfluoroalkyl acid uptake in lettuce (*Lactuca sativa*) and strawberry (*Fragaria ananassa*) irrigated with reclaimed water. Environ. Sci. Technol. 48, 14361–14368.
Bossi, R., Riget, F.F., Dietz, R., Sonne, C., Fauser, P., Dam, M., Vorkamp, K., 2005. Preliminary screening of perfluorooctane sulfonate (PFOS) and other fluorochemicals in fish, birds and marine mammals from Greenland and the Faroe Islands. Environ. Pollut. 136, 323–329.
Brambilla, G., D'Hollander, W., Oliaei, F., Stahl, T., Weber, R., 2015. Pathways and factors for food safety and food security at PFOS contaminated sites within a problem

based learning approach. Chemosphere 129, 192–202.
Buck, R.C., Franklin, J., Berger, U., Conder, J.M., Cousins, I.T., De Voogt, P., Jensen, A.A., Kannan, K., Mabury, S.A., van Leeuwen, S.P., 2011. Perfluoroalkyl and polyfluoroalkyl substances in the environment: terminology, classification, and origins. Integr. Environ. Assess. Manag. 7, 513–541.
Busch, J., Ahrens, L., Sturm, R., Ebinghaus, R., 2010. Polyfluoroalkyl compounds in landfill leachates. Environ. Pollut. 158, 1467–1471.
C8 Science Panel. The Science Panel Website. 〈http://www.c8sciencepanel.org/index.html〉.
Cai, M., Zhao, Z., Yin, Z., Ahrens, L., Huang, P., Cai, M., Yang, H., He, J., Sturm, R., Ebinghaus, R., 2011. Occurrence of perfluoroalkyl compounds in surface waters from the North Pacific to the Arctic Ocean. Environ. Sci. Technol. 46, 661–668.
Calafat, A.M., Needham, L.L., Kuklenyik, Z., Reidy, J.A., Tully, J.S., Aguilar-Villalobos, M., Naeher, L.P., 2006. Perfluorinated chemicals in selected residents of the American continent. Chemosphere 63, 490–496.
Chen, Y., Zhang, Y., Wang, Y., Sun, C., Zhang, C., 2013. Synthesis, characterization, and self-assembly of amphiphilic fluorinated gradient copolymer. J. Appl. Polym. Sci. 127, 1485–1492.
Chen, Y.M., Guo, L.H., 2009. Fluorescence study on site-specific binding of perfluoroalkyl acids to human serum albumin. Arch. Toxicol. 83, 255–261.
Christensen, K.Y., Raymond, M., Blackowicz, M., Liu, Y., Thompson, B.A., Anderson, H.A., Turyk, M., 2017. Perfluoroalkyl substances and fish consumption. Environ. Res. 154, 145–151.
Conder, J.M., Hoke, R.A., Wolf, W.D., Russell, M.H., Buck, R.C., 2008. Are PFCAs bioaccumulative? A critical review and comparison with regulatory criteria and persistent lipophilic compounds. Environ. Sci. Technol. 42, 995–1003.
Cornelis, C., D'Hollander, W., Roosens, L., Covaci, A., Smolders, R., Van Den Heuvel, R., Govarts, E., Van Campenhout, K., Reynders, H., Bervoets, L., 2012. First assessment of population exposure to perfluorinated compounds in Flanders, Belgium. Chemosphere 86, 308–314.
D'Hollander, W., de Voogt, P., De Coen, W., Bervoets, L., 2010a. Perfluorinated substances in human food and other sources of human exposure. Rev. Environ. Contam. Toxicol. 208, 179–215.
D'Hollander, W., Herzke, D., Huber, S., Hajslova, J., Pulkrabova, J., Brambilla, G., De Filippis, S.P., Bervoets, L., de Voogt, P., 2015. Occurrence of perfluorinated alkylated substances in cereals, salt, sweets and fruit items collected in four European countries. Chemosphere 129, 179–185.
D'Hollander, W., Roosens, L., Covaci, A., Cornelis, C., Reynders, H., Van Campenhout, K., de Voogt, P., Bervoets, L., 2010b. Brominated flame retardants and perfluorinated compounds in indoor dust from homes and offices in Flanders, Belgium. Chemosphere 81, 478–487.
De Voogt, P., Sáez, M., 2006. Analytical chemistry of perfluoroalkylated substances. TrAC Trends Anal. Chem. 25, 326–342.
Toxicological Effects of Perfluoroalkyl and Polyfluoroalkyl Substances. In: DeWitt, J.C. (Ed.), Springer International Publishing: Cham.
EPA, 2016a. Health Effects Support Document for Perfluorooctane Sulfonate (PFOS). EPA 822-R-16-002. 〈https://www.epa.gov/sites/production/files/2016-05/documents/hesd_pfos_final-plain.pdf〉.
EPA, 2016b. Health Effects Support Document for Perfluorooctanoic Acid (PFOA). EPA 822-R-16-003. 〈https://www.epa.gov/sites/production/files/2016-05/documents/pfoa_hesd_final_508.pdf〉.
European Food Safety Authority, 2008. Opinion of the Scientific panel on contaminants in the food chain on perfluorooctane sulfonate (PFOS), perfluorooctanoic acid (PFOA) and their salts. EFSA J. 1–131.
Eriksen, K.T., Raaschou-Nielsen, O., Sørensen, M., Roursgaard, M., Loft, S., Møller, P., 2010. Genotoxic potential of the perfluorinated chemicals PFOA, PFOS, PFBS, PFNA and PFHxA in human HepG2 cells. Mutat. Res. Toxicol. Environ. Mutagen. 700, 39–43.
Fan, H., Ducatman, A., Zhang, J., 2014. Perfluorocarbons and Gilbert syndrome (phenotype) in the C8 Health Study Population. Environ. Res. 135, 70–75.
Fang, X., Zhang, L., Feng, Y., Zhao, Y., Dai, J., 2008. Immunotoxic effects of perfluorononanoic acid on BALB/c mice. Toxicol. Sci. 105, 312–321.
Felizeter, S., McLachlan, M.S., De Voogt, P., 2014. Root Uptake and Translocation of Perfluorinated Alkyl Acids by Three Hydroponically Grown Crops. J. Agric. Food Chem. 62, 3334–3342.
Felizeter, S., McLachlan, M.S., de Voogt, P., 2012. Uptake of perfluorinated Alkyl acids by Hydroponically grown lettuce (*Lactuca sativa*). Environ. Sci. Technol. 46, 11735–11743.
Filipovic, M., Woldegiorgis, A., Norström, K., Bibi, M., Lindberg, M., Österås, A.H., 2015. Historical usage of aqueous film forming foam: a case study of the widespread distribution of perfluoroalkyl acids from a military airport to groundwater, lakes, soils and fish. Chemosphere 129, 39–45.
Fromme, H., Tittlemier, S.A., Völkel, W., Wilhelm, M., Twardella, D., 2009. Perfluorinated compounds–exposure assessment for the general population in western countries. Int. J. Hyg. Environ. Health 212, 239–270.
Fuchibe, K., Akiyama, T., 2006. Low-valent niobium-mediated double activation of CF/CH bonds: fluorene synthesis from o-arylated α, α, α-Trifluorotoluene derivatives. J. Am. Chem. Soc. 128, 1434–1435.
Gellrich, V., Stahl, T., Knepper, T.P., 2012. Behavior of perfluorinated compounds in soils during leaching experiments. Chemosphere 87, 1052–1056.
Giesy, J.P., Kannan, K., 2001. Global distribution of perfluorooctane sulfonate in wildlife. Environ. Sci. Technol. 35, 1339–1342.
Gobelius, L., Lewis, J., Ahrens, L., 2017. Plant uptake of per-and polyfluoroalkyl substances at a contaminated fire training facility to evaluate the phytoremediation potential of various plant species. Environ. Sci. Technol. 51, 12602–12610.
Harris, M.W., Birnbaum, L.S., 1989. Developmental toxicity of perfluorodecanoic acid in C57BL/6N mice. Toxicol. Sci. 12, 442–448.
Haukås, M., Berger, U., Hop, H., Gulliksen, B., Gabrielsen, G.W., 2007. Bioaccumulation of per-and polyfluorinated alkyl substances (PFAS) in selected species from the Barents Sea food web. Environ. Pollut. 148, 360–371.
Henderson, W.M., Smith, M.A., 2007. Perfluorooctanoic acid and perfluorononanoic acid in fetal and neonatal mice following in utero exposure to 8-2 fluorotelomer alcohol. Toxicol. Sci. 95, 452–461.
Herzke, D., Huber, S., Bervoets, L., D'Hollander, W., Hajslova, J., Pulkrabova, J., Brambilla, G., De Filippis, S.P., Klenow, S., Heinemeyer, G., 2013. Perfluorinated alkylated substances in vegetables collected in four European countries; occurrence and human exposure estimations. Environ. Sci. Pollut. Res. 20, 7930–7939.
Heuvel, J.P.V., Kuslikis, B.I., Peterson, R.E., 1992. Covalent binding of perfluorinated fatty acids to proteins in the plasma, liver and testes of rats. Chem. Biol. Interact. 82, 317–328.
Higgins, C.P., Luthy, R.G., 2006. Sorption of perfluorinated surfactants on sediments. Environ. Sci. Technol. 40, 7251–7256.
Hloušková, V., Lanková, D., Kalachová, K., Hrádková, P., Poustka, J., Hajšlová, J., Pulkrabová, J., 2013. Occurrence of brominated flame retardants and perfluoroalkyl substances in fish from the Czech aquatic ecosystem. Sci. Total Environ. 461, 88–98.
Holmström, K.E., Järnberg, U., Bignert, A., 2005. Temporal trends of PFOS and PFOA in guillemot eggs from the Baltic Sea, 1968–2003. Environ. Sci. Technol. 39, 80–84.
Hölzer, J., Midasch, O., Rauchfuss, K., Kraft, M., Reupert, R., Angerer, J., Kleeschulte, P., Marschall, N., Wilhelm, M., 2008. Biomonitoring of perfluorinated compounds in children and adults exposed to perfluorooctanoate-contaminated drinking water. Environ. Health Perspect. 116, 651.
Houde, M., De Silva, A.O., Muir, D.C., Letcher, R.J., 2011. Monitoring of perfluorinated compounds in aquatic biota: an updated review: PFASs in aquatic biota. Environ. Sci. Technol. 45, 7962–7973.
Hu, W.Y., Jones, P.D., DeCoen, W., King, L., Fraker, P., Newsted, J., Giesy, J.P., 2003. Alterations in cell membrane properties caused by perfluorinated compounds. Comp. Biochem. Physiol. Part C Toxicol. Pharmacol. 135, 77–88.
Huset, C.A., Chiaia, A.C., Barofsky, D.F., Jonkers, N., Kohler, H.P.E., Ort, C., Giger, W., Field, J.A., 2008. Occurrence and mass flows of fluorochemicals in the Glatt Valley watershed, Switzerland. Environ. Sci. Technol. 42, 6369–6377.
IARC, 2017. MONOGRAPHS – 110. 〈https://monographs.iarc.fr/list-of-classifications-volumes/〉.
Inoue, K., Okada, F., Ito, R., Kato, S., Sasaki, S., Nakajima, S., Uno, A., Saijo, Y., Sata, F., Yoshimura, Y., 2004. Perfluorooctane sulfonate (PFOS) and related perfluorinated compounds in human maternal and cord blood samples: assessment of PFOS exposure in a susceptible population during pregnancy. Environ. Health Perspect. 112, 1204–1207.
Jin, Y.H., Liu, W., Sato, I., Nakayama, S.F., Sasaki, K., Saito, N., Tsuda, S., 2009. PFOS and PFOA in environmental and tap water in China. Chemosphere 77, 605–611.
Jones, P.D., Hu, W., De Coen, W., Newsted, J.L., Giesy, J.P., 2003. Binding of perfluorinated fatty acids to serum proteins. Environ. Toxicol. Chem. 22, 2639–2649.
Kannan, K., Koistinen, J., Beckmen, K., Evans, T., Gorzelany, J.F., Hansen, K.J., Jones, P.D., Helle, E., Nyman, M., Giesy, J.P., 2001. Accumulation of perfluorooctane sulfonate in marine mammals. Environ. Sci. Technol. 35, 1593–1598.
Kannan, K., Tao, L., Sinclair, E., Pastva, S.D., Jude, D.J., Giesy, J.P., 2005. Perfluorinated compounds in aquatic organisms at various trophic levels in a Great Lakes food chain. Arch. Environ. Contam. Toxicol. 48, 559–566.
Kissa, E., 1994. Fluorinated Surfactants: Synthesis, Properties, Applications. Marcel Dekker Inc, pp. 50.
Klaunig, J.E., Shinohara, M., Iwai, H., Chengelis, C.P., Kirkpatrick, J.B., Wang, Z., Bruner, R.H., 2014. Evaluation of the chronic toxicity and carcinogenicity of perfluorohexanoic acid (PFHxA) in Sprague-Dawley rats. Toxicol. Pathol. 43, 209–220.
Klenow, S., Heinemeyer, G., Brambilla, G., Dellatte, E., Herzke, D., de Voogt, P., 2013. Dietary exposure to selected perfluoroalkyl acids (PFAAs) in four European regions. Food Addit. Contam. Part A 30, 2141–2151.
Korucu, M.K., Gedik, K., Weber, R., Karademir, A., Kurt-Karakus, P.B., 2015. Inventory development for perfluorooctane sulfonic acid (PFOS) in Turkey: challenges to control chemicals in articles and products. Environ. Sci. Pollut. Res. 22, 14537–14545.
Kowalczyk, J., Ehlers, S., Oberhausen, A., Tischer, M., Fürst, P., Schafft, H., Lahrssen-Wiederholt, M., 2013. Absorption, distribution, and milk secretion of the perfluoroalkyl acids PFBS, PFHxS, PFOS, and PFOA by dairy cows fed naturally contaminated feed. J. Agric. Food Chem. 61, 2903–2912.
Krafft, M.P., Riess, J.G., 2015. Selected physicochemical aspects of poly-and perfluoroalkylated substances relevant to performance, environment and sustainability-Part one. Chemosphere 129, 4–19.
Krippner, J., Brunn, H., Falk, S., Georgii, S., Schubert, S., Stahl, T., 2014. Effects of chain length and pH on the uptake and distribution of perfluoroalkyl substances in maize (*Zea mays*). Chemosphere 94, 85–90.
Krippner, J., Falk, S., Brunn, H., Georgii, S., Schubert, S., Stahl, T., 2015. Accumulation potentials of perfluoroalkyl carboxylic acids (PFCAs) and perfluoroalkyl sulfonic acids (PFSAs) in maize (*Zea mays*). J. Agric. Food Chem. 63, 3646–3653.
Kudo, N., Kawashima, Y., 2003. Toxicity and toxicokinetics of perfluorooctanoic acid in humans and animals. J. Toxicol. Sci. 28, 49–57.
Lassen, C., Jensen, A.A., Potrykus, A., Christensen, F., Kjølholt, J., Nyander Jeppesen, C., Mikkelsen, S.H., Innanen, S., 2012. Survey of PFOS, PFOA and other perfluoroalkyl and polyfluoroalkyl substances. Report Danish Environmental Protection Agency.
Latała, A., Nędzi, M., Stepnowski, P., 2009. Acute toxicity assessment of perfluorinated carboxylic acids towards the Baltic microalgae. Environ. Toxicol. Pharmacol. 28, 167–171.
Lechner, M., Knapp, H., 2011. Carryover of perfluorooctanoic acid (PFOA) and perfluorooctane sulfonate (PFOS) from soil to plant and distribution to the different plant compartments studied in cultures of carrots (*Daucus carota* ssp. sativus),

potatoes (*Solanum tuberosum*), and cucumbers (*Cucumis sativus*). J. Agric. Food Chem. 59, 11011–11018.
Li, Y., Oliver, D.P., Kookana, R.S., 2018. A critical analysis of published data to discern the role of soil and sediment properties in determining sorption of per and polyfluoroalkyl substances (PFASs). Sci. Total Environ. 628, 110–120.
Lindim, C., Van Gils, J., Cousins, I.T., 2016. Europe-wide estuarine export and surface water concentrations of PFOS and PFOA. Water Res. 103, 124–132.
Liu, H., Zhang, H., Cui, R., Guo, X., Wang, D., Dai, J., 2016. Activation of peroxisome proliferator-activated receptor α ameliorates perfluorododecanoic acid-induced production of reactive oxygen species in rat liver. Arch. Toxicol. 90, 1383–1397.
Liu, Z., Lu, Y., Shi, Y., Wang, P., Jones, K., Sweetman, A.J., Su, C., 2017. Crop bioaccumulation and human exposure of perfluoroalkyl acids through multi-media transport from a mega fluorochemical industrial park, China. Environ. Int 106, 37–47.
Llorca, M., Farré, M., Picó, Y., Müller, J., Knepper, T.P., Barceló, D., 2012. Analysis of perfluoroalkyl substances in waters from Germany and Spain. Sci. Total Environ. 431, 139–150.
Loccisano, A.E., Campbell, J.L., Butenhoff, J.L., Andersen, M.E., Clewell, H.J., 2012. Evaluation of placental and lactational pharmacokinetics of PFOA and PFOS in the pregnant, lactating, fetal and neonatal rat using a physiologically based pharmacokinetic model. Reprod. Toxicol. 33, 468–490.
Loos, R., Gawlik, B.M., Locoro, G., Rimaviciute, E., Contini, S., Bidoglio, G., 2009. EU-wide survey of polar organic persistent pollutants in European river waters. Environ. Pollut. 157, 561–568.
Manzetti, S., van der Spoel, E.R., van der Spoel, D., 2014. Chemical properties, environmental fate, and degradation of Seven classes of pollutants. Chem. Res. Toxicol. 27, 713–737.
Martin, J.W., Mabury, S.A., Solomon, K.R., Muir, D.C., 2003a. Bioconcentration and tissue distribution of perfluorinated acids in rainbow trout (*Oncorhynchus mykiss*). Environ. Toxicol. Chem. 22, 196–204.
Martin, J.W., Mabury, S.A., Solomon, K.R., Muir, D.C., 2003b. Dietary accumulation of perfluorinated acids in juvenile rainbow trout (*Oncorhynchus mykiss*). Environ. Toxicol. Chem. 22, 189–195.
Moody, C.A., Field, J.A., 1999. Determination of perfluorocarboxylates in groundwater impacted by fire-fighting activity. Environ. Sci. Technol. 33, 2800–2806.
Müller, C.E., LeFevre, G.H., Timofte, A.E., Hussain, F.A., Sattely, E.S., Luthy, R.G., 2016. Competing mechanisms for perfluoroalkyl acid accumulation in plants revealed using an Arabidopsis model system. Environ. Toxicol. Chem. 35, 1138–1147.
Navarro, I., de la Torre, A., Sanz, P., Porcel, M.Á., Pro, J., Carbonell, G., de los Ángeles Martínez, M., 2017. Uptake of perfluoroalkyl substances and halogenated flame retardants by crop plants grown in biosolids-amended soils. Environ. Res. 152, 199–206.
O'Hagan, D., 2008. Understanding organofluorine chemistry. An introduction to the C–F bond. Chem. Soc. Rev. 37, 308–319.
Parsons, J.R., Sáez, M., Dolfing, J., de Voogt, P., 2008. Biodegradation of perfluorinated compounds. Rev. Environ. Contam. Toxicol. 196, 53–71.
Paul, A.G., Jones, K.C., Sweetman, A.J., 2008. A first global production, emission, and environmental inventory for perfluorooctane sulfonate. Environ. Sci. Technol. 43, 386–392.
Qin, P., Liu, R., Pan, X., Fang, X., Mou, Y., 2010. Impact of carbon chain length on binding of perfluoroalkyl acids to bovine serum albumin determined by spectroscopic methods. J. Agric. Food Chem. 58, 5561–5567.
Rankin, K., Mabury, S.A., Jenkins, T.M., Washington, J.W., 2016. A North American and global survey of perfluoroalkyl substances in surface soils: distribution patterns and mode of occurrence. Chemosphere 161, 333–341.
Ren, X.M., Zhang, Y.F., Guo, L.H., Qin, Z.F., Lv, Q.Y., Zhang, L.Y., 2015. Structure–activity relations in binding of perfluoroalkyl compounds to human thyroid hormone T3 receptor. Arch. Toxicol. 89, 233–242.
Ritter, S.K., 2010. Fluorochemicals go short. Chem. Eng. News 88, 12–17. https://doi.org/10.1021/cen-v088n005.p012.
Rosen, M.B., Das, K.P., Wood, C.R., Wolf, C.J., Abbott, B.D., Lau, C., 2013. Evaluation of perfluoroalkyl acid activity using primary mouse and human hepatocytes. Toxicology 308, 129–137.
Seacat, A.M., Thomford, P.J., Hansen, K.J., Clemen, L.A., Eldridge, S.R., Elcombe, C.R., Butenhoff, J.L., 2003. Sub-chronic dietary toxicity of potassium perfluorooctanesulfonate in rats. Toxicology 183, 117–131.
Seacat, A.M., Thomford, P.J., Hansen, K.J., Olsen, G.W., Case, M.T., Butenhoff, J.L., 2002. Subchronic toxicity studies on perfluorooctanesulfonate potassium salt in cynomolgus monkeys. Toxicol. Sci. 68, 249–264.
Schenker, U., Scheringer, M., Macleod, M., Martin, J.W., Cousins, I.T., Hungerbühler, K., 2008. Contribution of volatile precursor substances to the flux of perfluorooctanoate to the Arctic. Environ. Sci. Technol. 42, 3710–3716.
Sepulvado, J.G., Blaine, A.C., Hundal, L.S., Higgins, C.P., 2011. Occurrence and fate of perfluorochemicals in soil following the land application of municipal biosolids. Environ. Sci. Technol. 45, 8106–8112.
Simonich, S.L., Hites, R.A., 1995. Organic pollutant accumulation in vegetation. Environ. Sci. Technol. 29, 2905–2914.
Skutlarek, D., Exner, M., Farber, H., 2006. Perfluorinated surfactants in surface and drinking waters. Environ. Sci. Pollut. Res. Int. 13, 299.
So, M., Miyake, Y., Yeung, W., Ho, Y., Taniyasu, S., Rostkowski, P., Yamashita, N., Zhou, B., Shi, X., Wang, J., 2007. Perfluorinated compounds in the Pearl river and Yangtze river of China. Chemosphere 68, 2085–2095.
So, M., Taniyasu, S., Yamashita, N., Giesy, J., Zheng, J., Fang, Z., Im, S., Lam, P.K., 2004. Perfluorinated compounds in coastal waters of Hong Kong, South China, and Korea. Environ. Sci. Technol. 38, 4056–4063.
Stahl, T., Heyn, J., Thiele, H., Hüther, J., Failing, K., Georgii, S., Brunn, H., 2009. Carryover of perfluorooctanoic acid (PFOA) and perfluorooctane sulfonate (PFOS) from soil to plants. Arch. Environ. Contam. Toxicol. 57, 289–298.
Stahl, T., Riebe, R.A., Falk, S., Failing, K., Brunn, H., 2013. Long-term lysimeter experiment to investigate the leaching of perfluoroalkyl substances (PFASs) and the carry-over from soil to plants: results of a pilot study. J. Agric. Food Chem. 61, 1784–1793.
Takahashi, M., Ishida, S., Hirata-Koizumi, M., Ono, A., Hirose, A., 2014. Repeated dose and reproductive/developmental toxicity of perfluoroundecanoic acid in rats. J. Toxicol. Sci. 39, 97–108.
Tian, Y., Yao, Y., Chang, S., Zhao, Z., Zhao, Y., Yuan, X., Sun, H., 2018. Occurrence and phase distribution of neutral and ionizable per-and polyfluoroalkyl substances (PFASs) in the atmosphere and plant leaves around landfills: a case study in Tianjin, China. Environ. Sci. Technol. 52 (3), 1301–1310.
Tittlemier, S.A., Pepper, K., Seymour, C., Moisey, J., Bronson, R., Cao, X.L., Dabeka, R.W., 2007. Dietary exposure of Canadians to perfluorinated carboxylates and perfluorooctane sulfonate via consumption of meat, fish, fast foods, and food items prepared in their packaging. J. Agric. Food Chem. 55, 3203–3210.
Upham, B.L., Park, J.S., Babica, P., Sovadinova, I., Rummel, A.M., Trosko, J.E., Hirose, A., Hasegawa, R., Kanno, J., Sai, K., 2009. Structure-activity-dependent regulation of cell communication by perfluorinated fatty acids using in vivo and in vitro model systems. Environ. Health Perspect. 117, 545.
Valsecchi, S., Rusconi, M., Mazzoni, M., Viviano, G., Pagnotta, R., Zaghi, C., Serrini, G., Polesello, S., 2015. Occurrence and sources of perfluoroalkyl acids in Italian river basins. Chemosphere 129, 126–134.
Vierke, L., Staude, C., Biegel-Engler, A., Drost, W., Schulte, C., 2012. Perfluorooctanoic acid (PFOA)—main concerns and regulatory developments in Europe from an environmental point of view. Environ. Sci. Eur. 24, 16.
Vongphachan, V., Cassone, C.G., Wu, D., Chiu, S., Crump, D., Kennedy, S.W., 2011. Effects of perfluoroalkyl compounds (PFASs) on mRNA expression levels of thyroid hormone-responsive genes in primary cultures of avian neuronal cells. Toxicol. Sci. 120, 392–402.
Wang, Z., Cousins, I.T., Scheringer, M., Hungerbuehler, K., 2015. Hazard assessment of fluorinated alternatives to long-chain perfluoroalkyl acids (PFAAs) and their precursors: status quo, ongoing challenges and possible solutions. Environ. Int. 75, 172–179.
Washington, J.W., Yoo, H., Ellington, J.J., Jenkins, T.M., Libelo, E.L., 2010. Concentrations, distribution, and persistence of perfluoroalkylates in sludge-applied soils near Decatur, Alabama, USA. Environ. Sci. Technol. 44, 8390–8396.
Weber, R., Watson, A., Forter, M., Oliaei, F., 2011. Persistent organic pollutants and landfills-a review of past experiences and future challenges. Waste Manag. Res. 29, 107–121.
Wen, B., Wu, Y., Zhang, H., Liu, Y., Hu, X., Huang, H., Zhang, S., 2016. The roles of protein and lipid in the accumulation and distribution of perfluorooctane sulfonate (PFOS) and perfluorooctanoate (PFOA) in plants grown in biosolids-amended soils. Environ. Pollut. 216, 682–688.
Wen, B., Zhang, H., Li, L., Hu, X., Liu, Y., Shan, X.Q., Zhang, S., 2015. Bioavailability of perfluorooctane sulfonate (PFOS) and perfluorooctanoic acid (PFOA) in biosolids-amended soils to earthworms (*Eisenia fetida*). Chemosphere 118, 361–366.
Wen, B., Li, L., Liu, Y., Zhang, H., Hu, X., Shan, X., Zhang, S., 2013. Mechanistic studies of perfluorooctane sulfonate, perfluorooctanoic acid uptake by maize (*Zea mays* L. cv. TY2). Plant Soil 370, 345–354.
Wen, B., Li, L., Zhang, H., Ma, Y., Shan, X.Q., Zhang, S., 2014. Field study on the uptake and translocation of perfluoroalkyl acids (PFAAs) by wheat (*Triticum aestivum* L.) grown in biosolids-amended soils. Environ. Pollut. 184, 547–554.
Wilhelm, M., Kraft, M., Rauchfuss, K., Hölzer, J., 2008. Assessment and management of the first German case of a contamination with Perfluorinated compounds (PFC) in the Region Sauerland, North Rhine-Westphalia. J. Toxicol. Env. Health, Part A 71, 725–733.
Wilhelm, M., Hölzer, J., Dobler, L., Rauchfuss, K., Midasch, O., Kraft, M., Angerer, J., Wiesmüller, G., 2009. Preliminary observations on perfluorinated compounds in plasma samples (1977–2004) of young German adults from an area with perfluorooctanoate-contaminated drinking water. Int. J. Hyg. Environ. Health 212, 142–145.
Winkens, K., Vestergren, R., Berger, U., Cousins, I.T., 2017. Early life exposure to per-and polyfluoroalkyl substances (PFASs): a critical review. Emerg. Contam. 3, 55–68.
Wolf, C.J., Fenton, S.E., Schmid, J.E., Calafat, A.M., Kuklenyik, Z., Bryant, X.A., Thibodeaux, J., Das, K.P., White, S.S., Lau, C.S., 2007. Developmental toxicity of perfluorooctanoic acid in the CD-1 mouse after cross-foster and restricted gestational exposures. Toxicol. Sci. 95, 462–473.
Wolf, C.J., Schmid, J.E., Lau, C., Abbott, B.D., 2012. Activation of mouse and human peroxisome proliferator-activated receptor-alpha (PPARα) by perfluoroalkyl acids (PFAAs): Further investigation of C4–C12 compounds. Reprod. Toxicol. 33, 546–551.
Wolf, C.J., Zehr, R.D., Schmid, J.E., Lau, C., Abbott, B.D., 2010. Developmental effects of perfluorononanoic acid in the mouse are dependent on peroxisome proliferator-activated receptor-alpha. PPAR Res. 2010.
Xiao, F., Halbach, T.R., Simcik, M.F., Gulliver, J.S., 2012. Input characterization of perfluoroalkyl substances in wastewater treatment plants: source discrimination by exploratory data analysis. Water Res. 46, 3101–3109.
Yamashita, N., Kannan, K., Taniyasu, S., Horii, Y., Petrick, G., Gamo, T., 2005. A global survey of perfluorinated acids in oceans. Mar. Pollut. Bull. 51, 658–668.
Yin, T., Chen, H., Reinhard, M., Yi, X., He, Y., Gin, K.Y.H., 2017. Perfluoroalkyl and polyfluoroalkyl substances removal in a full-scale tropical constructed wetland system treating landfill leachate. Water Res. 125, 418–426.
Yoo, H., Washington, J.W., Jenkins, T.M., Ellington, J.J., 2011. Quantitative determination of perfluorochemicals and fluorotelomer alcohols in plants from biosolid-amended fields using LC/MS/MS and GC/MS. Environ. Sci. Technol. 45, 7985–7990.
You, C., Jia, C., Pan, G., 2010. Effect of salinity and sediment characteristics on the sorption and desorption of perfluorooctane sulfonate at sediment-water interface.

Environ. Pollut. 158, 1343–1347.
Zareitalabad, P., Siemens, J., Harmer, M., Amelung, W., 2013. Perfluorooctanoic acid (PFOA) and perfluorooctanesulfonic acid (PFOS) in surface waters, sediments, soils and wastewater - a review on concentrations and distribution coeffcients. Chemosphere 91, 725–732.
Zhang, H., Hou, J., Cui, R., Guo, X., Shi, Z., Yang, F., Dai, J., 2013. Phosphoproteome analysis reveals an important role for glycogen synthase kinase-3 in perfluorododecanoic acid-induced rat liver toxicity. Toxicol. Lett. 218, 61–69.
Zhao, H., Chen, C., Zhang, X., Chen, J., Quan, X., 2011. Phytotoxicity of PFOS and PFOA to *Brassica chinensis* in different Chinese soils. Ecotoxicol. Environ. Saf. 74, 1343–1347.
Zhao, H., Guan, Y., Zhang, G., Zhang, Z., Tan, F., Quan, X., Chen, J., 2013. Uptake of perfluorooctane sulfonate (PFOS) by wheat (*Triticum aestivum* L.) plant. Chemosphere 91, 139–144.
Zhao, H., Qu, B., Guan, Y., Jiang, J., Chen, X., 2016. Influence of salinity and temperature on uptake of perfluorinated carboxylic acids (PFCAs) by hydroponically grown wheat (*Triticum aestivum* L.). SpringerPlus 5, 541.
Zhao, L., Zhu, L., Zhao, S., Ma, X., 2016. Sequestration and bioavailability of perfluoroalkyl acids (PFAAs) in soils: implications for their underestimated risk. Sci. Total Environ. 572, 169–176.
Zhao, S., Fang, S., Zhu, L., Liu, L., Liu, Z., Zhang, Y., 2014. Mutual impacts of wheat (*Triticum aestivum* L.) and earthworms (*Eisenia fetida*) on the bioavailability of perfluoroalkyl substances (PFASs) in soil. Environ. Pollut. 184, 495–501.
Zhao, S., Zhu, L., Liu, L., Liu, Z., Zhang, Y., 2013. Bioaccumulation of perfluoroalkyl carboxylates (PFASAs) and perfluoroalkane sulfonates (PFSAs) by earthworms (*Eisenia fetida*) in soil. Environ. Pollut. 2013 (179), 45–52.
Zhao, Y., Wong, C., Wong, M., 2012. Environmental contamination, human exposure and body loadings of perfluorooctane sulfonate (PFOS), focusing on Asian countries. Chemosphere 89, 355–368.